

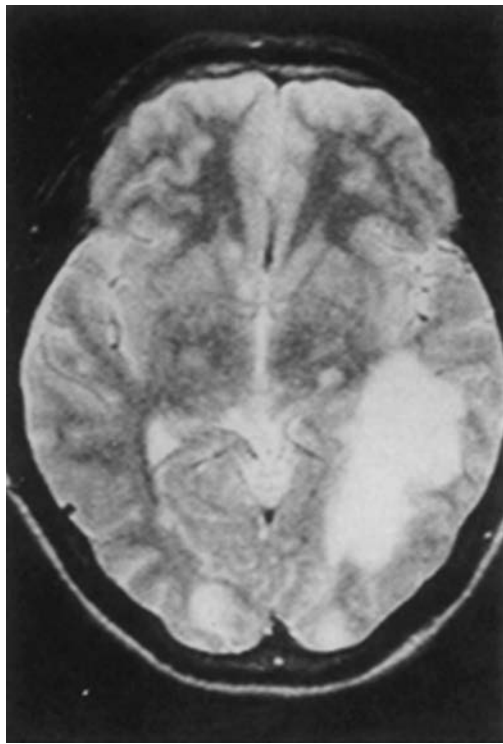


FIGURE 223-5 Brain MRI of amebic meningoencephalitis due to *Balamuthia mandrillaris*. A large lesion in the parieto-occipital lobe and other smaller lesions are seen. (Courtesy of the Department of Radiology, UCSD Medical Center, San Diego.)

granulomatous amebic encephalitis identified by the California Encephalitis Project include young age, immunocompromising conditions, and Hispanic ethnicity. The infection likely starts with percutaneous or mucous membrane exposure and then spreads hematogenously to the brain and other organs—a pattern that explains the risk for transmission through organ transplantation. In 2009–2010, two clusters of organ transplant–transmitted *B. mandrillaris* infections were detected by recognition of severe unexpected illness in multiple recipients from the same donor after an incubation period of 17–24 days.

Frequently, *Balamuthia* affects immunocompetent individuals, in whom the course is typically subacute, with focal neurologic signs, fever, seizures, and headaches leading to death within 1 week to several months after onset. Skin lesions may occur on the face, trunk, or extremities. In addition to dust inhalation, inoculation of trophozoites or cysts from stagnant water may occur through open wounds or mucous membranes. Diagnosis relies on examination of CSF, which reveals mononuclear or neutrophilic pleocytosis, elevated protein levels, and normal to low glucose concentrations. Amebae are rarely isolated from CSF. Multiple hypodense lesions are usually detected with imaging studies (Fig. 223-5). Fluorescent antibody and PCR assays are available from the CDC.

The five surviving patients in the United States have been treated with a variety of drugs, including pentamidine, flucytosine, sulfadiazine, and macrolides. The CDC recommends that miltefosine now be included, as for treatment of other free-living amebae. The differential diagnosis includes tuberculomas (Chap. 178) and neurocysticercosis (Chap. 235).

■ FURTHER READING

Amebiasis

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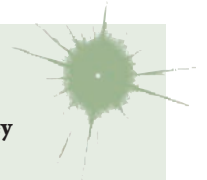
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Malaria

Nicholas J. White, Elizabeth A. Ashley



Humanity has but three great enemies: Fever, famine, and war; of these by far the greatest, by far the most terrible, is fever.

—William Osler, 1896

Malaria is a protozoan disease transmitted by the bite of infected female *Anopheles* mosquitoes. The most important of the parasitic diseases of humans, malaria is transmitted in 87 countries containing 3 billion people. In 2019, it was estimated that there were 229 million cases and 409,000 deaths (i.e., ~1100 deaths each day). Mortality rates decreased dramatically between 2000 and 2015 as a result of highly effective control programs in several countries, but since then, progress has reversed and estimated global case numbers have risen steadily. Malaria was eliminated from the United States, Canada, Europe, and Russia >50 years ago, but its prevalence rose in many parts of the tropics between 1970 and 2000. In response to this rise, there has been substantial investment aimed at increasing access to accurate diagnosis, effective treatments, and insecticide-treated bed nets. An increasing number of countries that had low malaria transmission are now targeting malaria elimination. This ambitious goal is threatened by increasing resistance to antimalarial drugs and insecticides.

Malaria remains today, as it has been for centuries, a heavy burden on tropical communities, a threat to nonendemic countries, and a danger to travelers.

ETIOLOGY AND PATHOGENESIS

Six species of the genus *Plasmodium* cause nearly all malarial infections in humans. These are *P. falciparum*, *P. vivax*, two morphologically identical sympatric species of *P. ovale* (*curtisi* and *wallikeri*), *P. malariae*, and—in Southeast Asia—the monkey malaria parasite *P. knowlesi* (Table 224-1). Occasionally humans are also infected with the monkey parasites *P. simium* (South America) and *P. cynomolgi* (Southeast Asia). While almost all deaths are caused by *falciparum* malaria, *P. knowlesi* and *P. vivax* can also cause severe illness. Human infection begins when a female anopheline mosquito inoculates plasmodial sporozoites from its salivary glands during a blood meal (Fig. 224-1). These microscopic motile forms of the malaria parasite are carried rapidly via the bloodstream to the liver, where they invade hepatic parenchymal cells and begin a period of asexual reproduction. By this amplification process (known as *intrahepatic* or *preerythrocytic schizogony*), a single sporozoite may produce from 10,000 to >30,000 daughter merozoites. These few swollen infected liver cells eventually burst, discharging motile

TABLE 224-1 Characteristics of *Plasmodium* Species Infecting Humans

CHARACTERISTIC	FINDING FOR INDICATED SPECIES				
	<i>P. FALCIPARUM</i>	<i>P. VIVAX</i>	<i>P. OVALE</i> ^a	<i>P. MALARIAE</i>	<i>P. KNOWLESI</i>
Duration of intrahepatic phase (days)	5.5	8	9	15	5.5
Number of merozoites released per infected hepatocyte	30,000	10,000	15,000	15,000	20,000
Approximate duration of erythrocytic cycle (hours)	48	48	50	72	24
Red cell preference	Younger cells (but can invade cells of all ages)	Reticulocytes and cells up to 2 weeks old	Reticulocytes	Older cells	Younger cells
Morphology	Usually only ring forms; banana-shaped gametocytes	Irregularly shaped large rings and trophozoites; enlarged erythrocytes; Schüffner's dots	Infected erythrocytes, enlarged and oval with tufted ends; Schüffner's dots	Band or rectangular forms of trophozoites common	Resembles <i>P. falciparum</i> (early trophozoites) or <i>P. malariae</i> (later trophozoites, including band forms)
Pigment color	Black	Yellow-brown	Dark brown	Brown-black	Dark brown
Ability to cause relapses	No	Yes	Yes	No	No

^aGenomic studies have revealed *P. ovale* to be two sympatric species: *P. ovale curtisi* and *P. ovale wallikeri*, which are morphologically very similar but may have different incubation periods and latencies.

merozoites into the bloodstream. The merozoites then invade red blood cells (RBCs) to become trophozoites and, in non-immune subjects, multiply six- to twentyfold every 48 h (*P. knowlesi*, 24 h; *P. malariae*, 72 h). When the parasites reach densities of ~50/μL of blood (~100 million parasites in total in the blood of an adult), the symptomatic stage of the infection begins. In *P. vivax* and *P. ovale* infections, a proportion of the intrahepatic forms do not divide immediately but remain inert for a period ranging from 2 weeks to ≥1 year. These dormant forms, or hypnozoites, are the cause of the relapses that characterize infection with these species.

Attachment of merozoites to erythrocytes is mediated via a complex interaction with several different binding ligands and specific erythrocyte surface receptors. *P. falciparum* merozoites bind via erythrocyte binding antigen 175 to glycophorin A and via EBL140 to glycophorin C. The other glycophorins (B and D) also contribute.

The merozoite reticulocyte-binding protein homologue 5 (PfRh5) plays a critical role binding to red cell basigin (CD147, EMMPRIN). *P. vivax* binds to receptors on developing erythrocytes. The Duffy blood-group antigen Fy^a or Fy^b plays an important role in invasion.

Most West Africans and people with origins in that region are the Duffy-negative FyFy phenotype and are generally resistant to *P. vivax* malaria. *P. knowlesi* also invades Duffy-positive human RBCs preferentially. During the first few hours of intraerythrocytic development, the small “ring forms” of the different malaria species appear similar under light microscopy. As the trophozoites enlarge, species-specific characteristics become evident, malaria pigment (hemozoin) becomes visible, and the parasite assumes an irregular or ameboid shape. By the end of the intraerythrocytic life cycle, the parasite has consumed two-thirds of the RBC's hemoglobin and has grown to occupy most of the cell. It is now called a schizont. Multiple nuclear divisions have taken place (schizogony or merogony). The infected RBC then ruptures to release 6–30 daughter merozoites, each potentially capable of invading a new RBC and repeating the cycle. The disease in human beings is caused by the direct effects of the asexual parasite—RBC invasion and destruction—and by the host's reaction. Some of the blood-stage parasites develop into morphologically distinct, longer-lived sexual forms (gametocytes) that can transmit malaria. In falciparum malaria, a delay of several asexual cycles precedes this switch to gametocytogenesis.

Female gametocytes typically outnumber males by 4:1.

After being ingested in the blood meal of a biting female anopheline mosquito, the male gametocyte exflagellates and divides rapidly into eight motile male gametes. These fuse with female gametocytes, undergoing two rounds of sexual division (meiosis) to form a zygote in the insect's midgut. This zygote matures into an ookinete, which penetrates and encysts in the mosquito's gut wall. The resulting oocyst expands by asexual division until it bursts to liberate myriad motile sporozoites, which then migrate in the hemolymph to the salivary gland of the mosquito to await inoculation into another human at the next feed, thus completing the life cycle.

EPIDEMIOLOGY

Malaria occurs throughout most of the tropical regions of the world (Fig. 224-2). *P. falciparum* predominates in Africa, New Guinea, and Hispaniola (i.e., the Dominican Republic and Haiti); *P. vivax* is more common in Central and

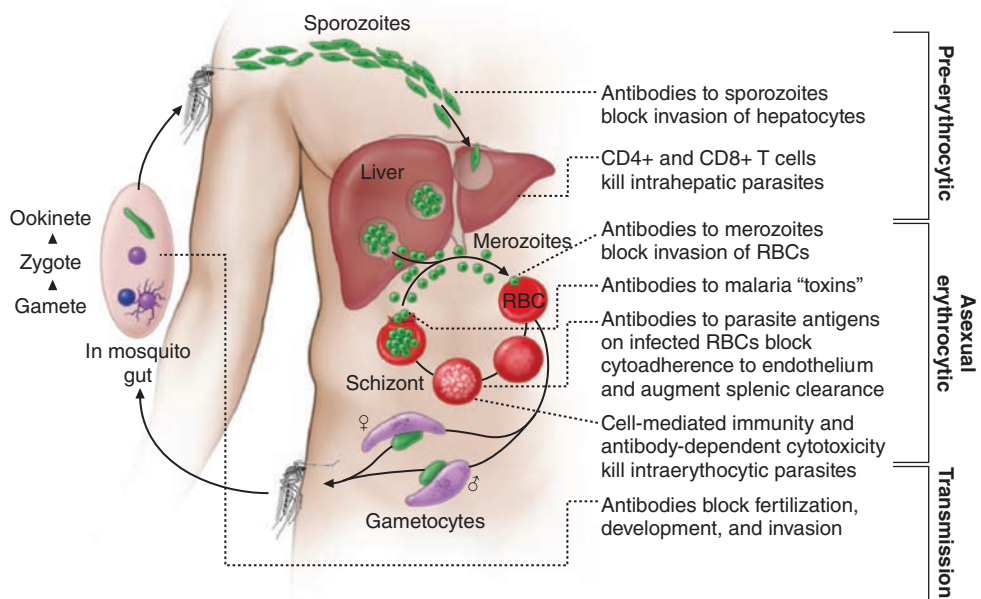


FIGURE 224-1 The malaria transmission cycle from mosquito to human and targets of immunity. In *P. vivax* and *P. ovale* infections some liver stage parasites remain dormant (“hypnozoites”), and awake weeks or months later to cause relapses. RBC, red blood cell.

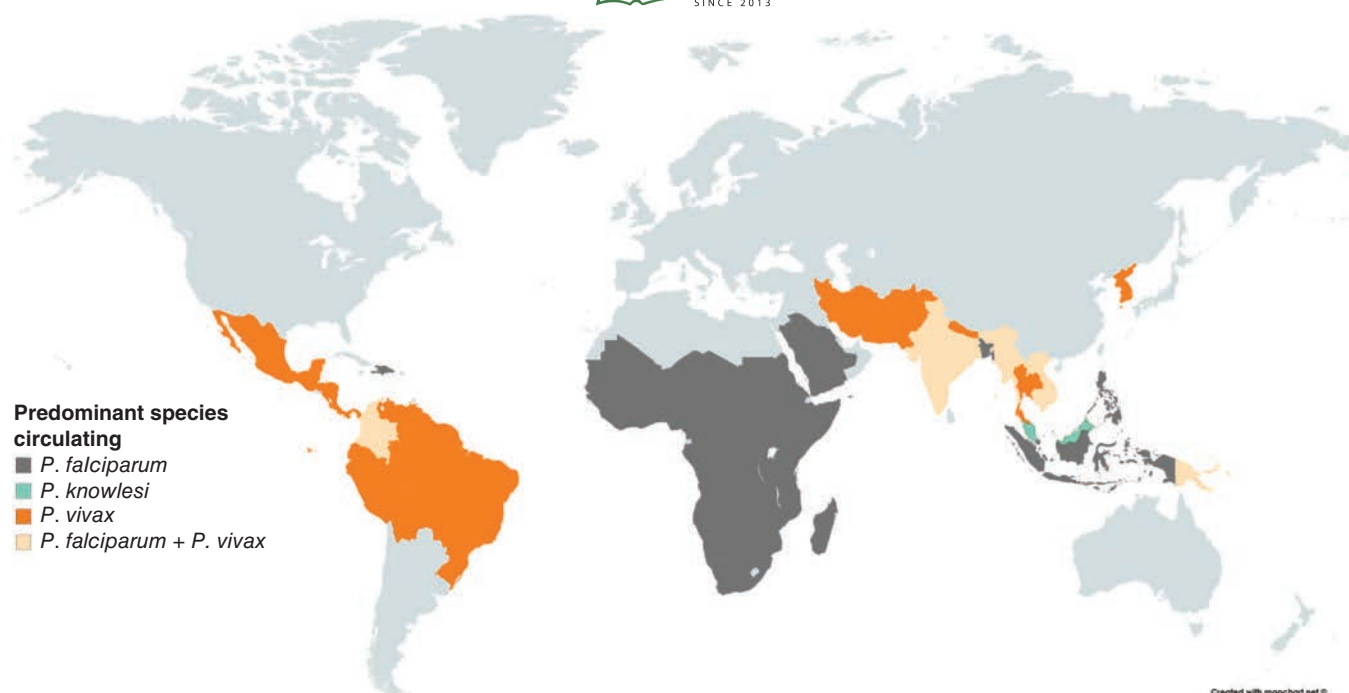


FIGURE 224-2 Malaria-endemic countries showing predominant *Plasmodium* species. *Plasmodium vivax* is common in the Horn of Africa and in Mauritania but relatively unusual elsewhere in the continent

South America and Southeast Asia. The prevalence of these two species is approximately equal on the Indian subcontinent and in Oceania. *P. malariae* is found in most endemic areas, especially throughout sub-Saharan Africa, but is much less common. *P. ovale* is relatively unusual outside of Africa and, where it is found, comprises <1% of isolates. *P. knowlesi* causes human infections commonly on the island of Borneo and, to a lesser extent, elsewhere in Southeast Asia, where the main hosts, long-tailed and pig-tailed macaques, are found.

The epidemiology of malaria is complex and may vary considerably even within relatively small geographic areas. Endemicity traditionally has been defined in terms of rates of microscopy-detected parasitemia or palpable spleens in children 2–9 years of age and has been classified as hypoendemic (<10%), mesoendemic (11–50%), hyperendemic (51–75%), and holoendemic (>75%). In holo- and hyperendemic areas (e.g., certain regions of tropical Africa or coastal New Guinea) where there is intense *P. falciparum* transmission, people may sustain one or more infectious mosquito bites per week and are infected repeatedly throughout their lives. In such settings, malaria morbidity and mortality are substantial during early childhood. Immunity against disease is hard won in these areas following repeated symptomatic infections in childhood, but, if the child survives, infections become increasingly likely to be asymptomatic. These asymptomatic older children and adults are a major source of malaria transmission. As control measures progress and urbanization expands, environmental conditions become less conducive to malaria transmission, and all age groups may lose protective immunity and become susceptible to illness. Constant, frequent, year-round infection is termed *stable transmission*. In areas where transmission is low, erratic, or focal, full protective immunity is not acquired, and symptomatic disease may occur at all ages. This situation usually exists in hypoendemic areas and is termed *unstable transmission*. Even in stable transmission areas, there is often an increased incidence of symptomatic malaria during the rainy season coinciding with increased mosquito breeding and transmission. Malaria can behave like an epidemic disease in some areas, particularly those with unstable malaria, such as northern India (the Punjab region), the Horn of Africa, Rwanda, Burundi, southern Africa, and Madagascar. Epidemics may occur when changes in environmental, economic, or social conditions (e.g., heavy rains following drought or migration—usually of refugees or workers—from a nonmalarious region to an area of high transmission) are compounded by failure to invest in national

programs or by a breakdown in malaria control and prevention services caused by war or civil disorder. The recent socioeconomic and political crisis in Venezuela has led to a resurgence of malaria. Epidemics often result in high mortality rates among all age groups. The principal determinants of the epidemiology of malaria are the number (density), the human-biting habits, and the longevity of the anopheline mosquito vectors. More than 100 of the >400 anopheline species can transmit malaria, but the ~40 species that do so commonly vary considerably in their efficiency as malaria vectors. More specifically, the transmission of malaria is directly proportional to the density of the vector, the square of the number of human bites per day per mosquito, and the tenth power of the probability of the mosquito's surviving for 1 day. Mosquito longevity is particularly important as a determinant of malaria transmissibility because the portion of the parasite's life cycle that takes place within the mosquito—from gametocyte ingestion to subsequent inoculation (*sporogony*)—lasts 8–30 days, depending on ambient temperature. In order to transmit malaria, the mosquito must therefore survive for >7 days. Sporogony is not completed at cooler temperatures—i.e., <16°C (<60.8°F) for *P. vivax* and <21°C (<69.8°F) for *P. falciparum*; thus, transmission does not occur below these temperatures or at high altitudes, although malaria outbreaks and transmission have occurred in the highlands (>1500 m) of eastern Africa, which were previously free of vectors. The most effective mosquito vectors of malaria are those, such as the *Anopheles gambiae* species complex in Africa, that are long-lived, occur in high densities in tropical climates, breed readily, and bite humans in preference to other animals. The entomologic inoculation rate (i.e., the number of sporozoite-positive mosquito bites per person per year) is the most common measure of malaria transmission and varies from <1 in some parts of Latin America and Southeast Asia to >300 in parts of tropical Africa.

PATHOPHYSIOLOGY

ERYTHROCYTE CHANGES

After invading an erythrocyte, the growing malarial parasite progressively consumes and degrades intracellular proteins, principally hemoglobin. The potentially toxic heme is detoxified by lipid-mediated crystallization to biologically inert hemozoin (malaria pigment). The parasite also alters the RBC membrane by changing its transport properties, exposing

cryptic surface antigens, and inserting new parasite-derived proteins. The RBC becomes more irregular in shape, more antigenic, and less deformable.

In *P. falciparum* infections, membrane protuberances appear on the erythrocyte's surface 12–15 h after cell invasion. These “knobs” extrude a high-molecular-weight, antigenically variant, strain-specific erythrocyte membrane adhesive protein (PfEMP1) that mediates attachment to receptors on venular and capillary endothelium (cytoadherence). Several vascular receptors have been identified; intercellular adhesion molecule 1 and endothelial protein C receptor are important in the brain, chondroitin sulfate B predominates in the placenta, and CD36 binds parasitized RBCs in most other organs. Erythrocytes containing more mature parasites stick inside and eventually block capillaries and venules. These infected RBCs may also adhere to uninfected RBCs (to form rosettes) and to other parasitized erythrocytes (agglutination). The processes of cytoadherence, rosetting, and agglutination are central to the pathogenesis of falciparum malaria. They result in the sequestration of infected RBCs in vital organs (particularly the brain), where they interfere with microcirculatory flow and metabolism. Sequestered parasites continue to develop out of reach of the principal host defense mechanism: splenic processing and filtration. As a consequence, only the younger ring forms of the asexual parasites circulate in the peripheral blood in falciparum malaria, and the level of peripheral parasitemia variably underestimates the true number of parasites within the body. In severe malaria, uninfected erythrocytes also become less deformable, which compromises their passage through the partially obstructed capillaries and venules and shortens their survival.

In the other human malarias, significant sequestration does not occur, and all stages of the parasite's development are evident on peripheral-blood smears. *P. vivax* and *P. ovale* show a marked predilection for young RBCs and *P. malariae* for old cells; these species produce a level of parasitemia that seldom exceeds 2%. In contrast, *P. falciparum* can invade erythrocytes of all ages and may be associated with very high parasite densities. Dangerously high parasite densities may also occur in *P. knowlesi* infections, with rapid increases as a result of the shorter (24-h) asexual life cycle.

■ HOST RESPONSE

Initially, the host responds to malaria infection by activating nonspecific defense mechanisms. Splenic immunologic and filtrative clearance functions are augmented, and the removal of both parasitized and uninfected erythrocytes is accelerated. The spleen also removes damaged ring-form parasites (a process known as “pitting”) from within the red cell and returns the once-infected cells back to the circulation, where their survival is shortened. The parasitized cells escaping splenic removal are destroyed when the schizont ruptures. The material released induces monocyte/macrophage activation and the release of proinflammatory cytokines, which cause fever and other pathologic effects. Temperatures of $\geq 40^{\circ}\text{C}$ ($\geq 104^{\circ}\text{F}$) damage mature parasites; in untreated infections, the effect of such temperatures is to further synchronize the parasitic cycle, with eventual production of the regular fever spikes and rigors that originally characterized the different malarias. These regular fever patterns (quotidian, daily; tertian, every 2 days; quartan, every 3 days) are seldom seen today as patients receive prompt and effective antimalarial treatment.

The geographic distributions of the thalassemias, sickle cell disease, hemoglobins C and E, hereditary ovalocytosis, and glucose-6-phosphate dehydrogenase (G6PD) deficiency closely resemble that of falciparum malaria before the introduction of control measures. This similarity suggests that these genetic disorders confer protection against death from falciparum malaria. HbA/S heterozygotes (sickle cell trait) have a sixfold reduction in the risk of dying from severe falciparum malaria and are correspondingly protected from the bacterial infections that complicate malaria. Hemoglobin S-containing RBCs impair parasite growth at low oxygen tensions, and *P. falciparum*-infected RBCs containing hemoglobin S or C exhibit reduced cytoadherence because of reduced surface presentation of the adhesin PfEMP1. Parasite multiplication in HbA/E heterozygotes is reduced at high parasite densities. In Melanesia, children with α -thalassemia have

more frequent malaria (both vivax and falciparum) in the early years of life, and this pattern of infection appears to protect them against severe disease. In Melanesian ovalocytosis, rigid erythrocytes resist merozoite invasion, and the intraerythrocytic milieu is hostile. G6PD deficiency provides some protection against severe *P. falciparum* infections but has a much stronger protective effect against *P. vivax* infections.

Nonspecific host defense mechanisms stop the infection's expansion, and the subsequent strain-specific immune response then controls the infection. Eventually, exposure to sufficient strains confers protection from high-level parasitemia and disease but not from infection. As a result of this state of infection without illness (premunition), asymptomatic parasitemia is very common among adults and older children living in regions with stable and intense transmission (i.e., holo- or hyperendemic areas) and also in parts of low-transmission areas. Parasitemia in asymptomatic infections fluctuates in density but often averages $\sim 5000/\text{mL}$ —just below the level of microscopy detection but sufficient to generate transmissible densities of gametocytes. Immunity is mainly specific for both the species and the strain of infecting malarial parasite. Both humoral immunity and cellular immunity are necessary for protection, but the mechanisms of each are incompletely understood (Fig. 224-1). Immune individuals have a polyclonal increase in serum levels of IgM, IgG, and IgA, although much of this antibody is unrelated to protection. Antibodies to a variety of parasite antigens presumably act in concert to limit in vivo replication of the parasite. In *P. falciparum* infections, the variant surface adhesin PfEMP1 is the most important of these antigens. Passive transfer of maternal antibody contributes to the partial protection of infants from severe malaria in the first months of life. This complex immunity to disease declines when a person lives outside an endemic area for several months or longer.

Several factors retard the development of cellular immunity to malaria. These factors include the absence of major histocompatibility antigens on the surface of infected RBCs, which precludes direct T cell recognition; malaria antigen-specific immune unresponsiveness; and the enormous strain diversity of malarial parasites, along with the ability of the parasites to express variant immunodominant antigens on the erythrocyte surface that change during the course of infection. Parasites may persist in the blood for months or years (or, in the case of *P. malariae*, for decades) if treatment is not given. The complexity of the immune response in malaria, the sophistication of the parasites' evasion mechanisms, and the lack of a good in vitro correlate with clinical immunity have all slowed progress toward an effective vaccine.

CLINICAL FEATURES

Malaria is a common cause of fever in tropical countries. Clinical diagnosis is notoriously unreliable. The first symptoms of malaria are nonspecific; the lack of a sense of well-being, headache, fatigue, abdominal discomfort, and muscle aches followed by fever are all similar to the symptoms of a minor viral illness. In some instances, a prominence of headache, chest pain, abdominal pain, cough, arthralgia, myalgia, or diarrhea may suggest another diagnosis. Although headache may be severe in malaria, the neck stiffness and photophobia seen in meningitis do not occur. While myalgia may be prominent, it is not usually as severe as in dengue fever, and the muscles are not tender as in leptospirosis or typhus. Nausea, vomiting, and orthostatic hypotension are common. The classic malarial paroxysms, in which fever spikes, chills, and rigors occur at regular intervals, are unusual and at presentation suggest infection (often relapse) with *P. vivax* or *P. ovale*. The fever is usually irregular at first (that of falciparum malaria may never become regular). The temperature of nonimmune individuals and children often rises above 40°C (104°F), with accompanying tachycardia and sometimes delirium. Although childhood febrile convulsions may occur with any of the malarias, generalized seizures are associated specifically with falciparum malaria and may herald the development of encephalopathy (cerebral malaria). Many clinical abnormalities have been described in acute malaria, but most patients with uncomplicated infections have few abnormal physical findings other than fever, malaise, mild anemia, and (in some cases) a palpable spleen. Anemia is common among young children living in areas with

1724 stable transmission (e.g., much of West Africa), and increases in prevalence where resistance has compromised the efficacy of antimalarial drugs. Frequent vivax malaria relapse is an important cause of anemia in young children in some areas (e.g., on the island of New Guinea). In nonimmune individuals with acute malaria, the spleen takes several days to become palpable, but splenic enlargement is found in a high proportion of otherwise healthy individuals in malaria-endemic areas and reflects repeated infections. Slight enlargement of the liver is also common, particularly among young children. Mild jaundice is common among adults; it may develop in patients with otherwise uncomplicated malaria and usually resolves over 1–3 weeks. Malaria is not associated with a rash. Petechial hemorrhages in the skin or mucous membranes—features of viral hemorrhagic fevers and leptospirosis—develop only very rarely in severe falciparum malaria.

■ SEVERE FALCIPARUM MALARIA

Appropriately and promptly treated, uncomplicated falciparum malaria (i.e., that in which the patient can sit or stand unaided and can swallow medicines and food) carries a mortality rate of <0.1%. However, once vital-organ dysfunction occurs or the total proportion of erythrocytes infected increases to >2% (a level corresponding to >10¹² parasites in an adult), mortality risk rises steeply, depending on the immunity of the host. The major manifestations of severe falciparum malaria are shown in **Table 224-2**, and features indicating a poor prognosis are listed in **Table 224-3**.

? **Cerebral Malaria** Coma is a characteristic and ominous feature of falciparum malaria and, even with treatment, has been associated with death rates of ~20% among adults and 15% among children. Any obtundation, delirium, or abnormal behavior in falciparum malaria should be taken very seriously. The onset of coma may be gradual or sudden following a convulsion.

Cerebral malaria manifests as a diffuse symmetric encephalopathy; focal neurologic signs are unusual. Although some passive resistance to head flexion may be detected, signs of meningeal irritation are absent. The eyes may be divergent, and bruxism and a pout reflex are common, but other primitive reflexes are usually absent. The corneal reflexes are preserved, except in deep coma. Muscle tone may be either increased or decreased. The tendon reflexes are variable, and the plantar reflexes may be flexor or extensor; the abdominal and cremasteric reflexes are absent. Flexor or extensor posturing may be seen. On routine funduscopy, ~15% of patients have retinal hemorrhages; with pupillary dilation and indirect ophthalmoscopy, this figure increases to 30–40%. Other fundoscopic abnormalities (**Fig. 224-3**) include discrete spots of retinal opacification (30–60%), papilledema (8% among children, rare among adults), cotton wool spots (<5%), and decolorization of a retinal vessel or segment of vessel (occasional cases). Convulsions, which are usually generalized and often repeated, occur in ~10% of adults and up to 50% of children with cerebral malaria. More covert seizure activity is common, particularly among children, and may manifest as repetitive tonic-clonic eye movements or even hypersalivation. Whereas adults rarely (<3% of cases) suffer neurologic sequelae, ~10% of children surviving cerebral malaria—especially those with hypoglycemia, severe anemia, repeated seizures, and deep coma—have residual neurologic deficits when they regain consciousness; hemiplegia, cerebral palsy, cortical blindness, deafness, and impaired cognition may all occur. The majority of these deficits improve markedly or resolve completely within 6 months. However, the prevalence of some other deficits increases over time; ~10% of children surviving cerebral malaria have a persistent language deficit. There may also be deficits in learning, planning and executive functions, attention, memory, and nonverbal functioning. The incidence of epilepsy is increased and life expectancy decreased among these children.

Hypoglycemia Hypoglycemia, an important and common complication of severe malaria, is associated with a poor prognosis and is particularly problematic in children and pregnant women. Hypoglycemia in malaria results from both a failure of hepatic gluconeogenesis and an increase in the consumption of glucose by the host and, to a much lesser extent, the malaria parasites. This may be compounded by

TABLE 224-2 Manifestations of Severe Falciparum Malaria

SIGNS	MANIFESTATIONS
Major	
Unarousable coma/cerebral malaria	Failure to localize or respond appropriately to noxious stimuli; coma persisting for >30 min after generalized convulsion
Acidemia/acidosis	Arterial pH of <7.25, base deficit >8 meq/L, or plasma bicarbonate level of <15 mmol/L; venous lactate level of >5 mmol/L; manifests as labored deep breathing, often termed “respiratory distress”
Severe normochromic, normocytic anemia	Hematocrit of <15% or hemoglobin level of <50 g/L (<5 g/dL) with parasitemia level of >10,000/μL ^a
Renal failure	Serum or plasma creatinine level of >265 μmol/L (>3 mg/dL); urine output (24 h) of <400 mL for adults or <12 mL/kg for children; no improvement with rehydration ^b
Pulmonary edema/adult respiratory distress syndrome	Noncardiogenic pulmonary edema, often aggravated by overhydration
Hypoglycemia	Plasma glucose level of <2.2 mmol/L (<40 mg/dL)
Hypotension/shock	Systolic blood pressure of <50 mmHg in children 1–5 years or <80 mmHg in adults; core/skin temperature difference of >10°C; capillary refill >2 s
Bleeding/disseminated intravascular coagulation	Significant bleeding and hemorrhage from the gums, nose, and gastrointestinal tract and/or evidence of disseminated intravascular coagulation
Convulsions	More than two generalized seizures in 24 h; signs of continued seizure activity, sometimes subtle (e.g., tonic-clonic eye movements without limb or face movement)
Other	
Hemoglobinuria ^c	Macroscopic black, brown, or red urine; not associated with effects of oxidant drugs and red blood cell enzyme defects (such as G6PD deficiency)
Extreme weakness	Prostration; inability to sit unaided ^d
Hyperparasitemia	Parasitemia level of >5% in nonimmune patients (>10% in any patient)
Jaundice	Serum bilirubin level of >50 mmol/L (>3 mg/dL) if combined with a parasite density of 100,000/μL or other evidence of vital-organ dysfunction

^aThis is nonspecific and may include patients with chronic anemia; a parasitemia threshold of 100,000/μL is more specific for acute malarial anemia. ^bIn practice, urine output information is usually unavailable, so the plasma or serum creatinine alone is used. ^cHemoglobinuria may also occur in uncomplicated malaria and in patients with G6PD deficiency, particularly if they take oxidant drugs such as primaquine. ^dIn children who are normally able to sit.

Abbreviation: G6PD, glucose-6-phosphate dehydrogenase.

quinine, a powerful stimulant of pancreatic insulin secretion, which is still widely used for the treatment of both severe and uncomplicated falciparum malaria. Hyperinsulinemic hypoglycemia is especially troublesome in pregnant women receiving quinine treatment. In severe disease, the clinical diagnosis of hypoglycemia is difficult: the usual physical signs (sweating, gooseflesh, tachycardia) are absent, and the neurologic impairment caused by hypoglycemia cannot be distinguished from that caused by malaria.

Acidosis Acidosis, resulting from accumulation of organic acids, is an important cause of death from severe malaria, which in adults is often compounded by coexisting renal impairment. Hyperlactatemia commonly coexists with hypoglycemia. In children, ketoacidosis may contribute. Hydroxyphenyllactic acid, α-hydroxybutyric acid, and β-hydroxybutyric acid concentrations are elevated. Acidotic breathing, sometimes called “respiratory distress,” is a sign of poor prognosis. It is followed often by circulatory failure refractory to volume expansion or inotropic drug treatment and ultimately by respiratory arrest. Plasma concentrations of bicarbonate or lactate are the best biochemical prognosticators in severe malaria. Hypovolemia is not a major contributor to acidosis. Lactic acidosis is caused by the combination of anaerobic glycolysis in tissues where sequestered parasites interfere with

TABLE 224-3 Features Indicating a Poor Prognosis in Severe Falciparum Malaria

Clinical

Marked agitation
Hyperventilation (respiratory distress)
Low core temperature (<36.5°C; <97.7°F)
Bleeding
Deep coma
Repeated convulsions
Anuria
Shock

Laboratory

Biochemistry

Hypoglycemia (<2.2 mmol/L)
Hyperlactatemia (>5 mmol/L)
Acidemia (arterial pH <7.25, base deficit >8 meq/L, or serum HCO₃ <15 mmol/L)
Elevated serum creatinine (>265 μmol/L)
Elevated total bilirubin (>50 μmol/L)
Elevated liver enzymes (AST/ALT 3 times upper limit of normal)
Elevated muscle enzymes (CPK ↑, myoglobin ↑)
Elevated urate (>600 μmol/L)

Hematology

Leukocytosis (>12,000/μL)
Severe anemia (PCV <15%)
Coagulopathy
Low platelet count (<50,000/μL)
Prolonged prothrombin time (>3 s)
Prolonged partial thromboplastin time
Decreased fibrinogen (<200 mg/dL)

Parasitology

Hyperparasitemia
Increased mortality at >100,000/μL
High mortality at >500,000/μL
>20% of parasites identified as pigment-containing trophozoites and schizonts
>5% of neutrophils contain visible malaria pigment

Note: Increased risk of concomitant bacteremia in adults if >20% parasitemia.

Abbreviations: ALT, alanine aminotransferase; AST, aspartate aminotransferase; CPK, creatine phosphokinase; PCV, packed cell volume.

microcirculatory flow, lactate production by the parasites, and a failure of hepatic and renal lactate clearance.

Noncardiogenic Pulmonary Edema Adults with severe falciparum malaria may develop noncardiogenic pulmonary edema even after several days of antimalarial therapy. The pathogenesis of this variant of the adult respiratory distress syndrome is unclear. The mortality rate is >80%. Pulmonary edema can be precipitated by overly vigorous administration of IV fluid. Noncardiogenic pulmonary edema can also develop in otherwise uncomplicated vivax malaria, where recovery is usual.



Renal Impairment Acute kidney injury is common in severe falciparum malaria. The pathogenesis of renal failure is unclear but may be related to erythrocyte sequestration and agglutination interfering with renal microcirculatory flow and metabolism. Clinically and pathologically, this syndrome manifests as acute tubular necrosis. Acute renal failure may occur simultaneously with other vital-organ dysfunction (in which case the mortality risk is high) or may progress as other disease manifestations resolve. In survivors, urine flow resumes in a median of 4 days, and serum creatinine levels return to normal in a mean of 17 days (Chap. 310). Early dialysis or hemofiltration considerably improves the chances of survival, particularly in acute hypercatabolic renal failure. Oliguric renal failure is rare among children.

Hematologic Abnormalities Anemia results from accelerated RBC removal by the spleen, obligatory RBC destruction at parasite



FIGURE 224-3 The eye in cerebral malaria: perimacular whitening and pale-centered retinal hemorrhages. (Courtesy of N. Beare, T. Taylor, S. Harding, S. Lewallen, and M. Molyneux; with permission.)

schizogony, and ineffective erythropoiesis. In severe malaria, the deformability of both infected and uninfected RBCs is reduced. The degree of reduced deformability correlates with prognosis and with the development of anemia. Splenic clearance of all RBCs is increased. In non-immune individuals and in areas with unstable transmission, anemia can develop rapidly and transfusion is often required. A hemoglobin of ≤3g/dL on presentation is associated with increased mortality. Acute hemolytic anemia with massive hemoglobinuria (“blackwater fever”) may occur. Hemoglobinuria may contribute to renal injury. Some patients with blackwater fever have G6PD deficiency, but in the majority of cases, it is unclear why massive hemolysis has occurred. In non-immune patients sudden hemolysis may follow many days after artesunate treatment of hyperparasitemia, usually as a result of relatively synchronous loss of once-parasitized “pitted” RBCs. As a consequence of repeated malarial infections, children in high-transmission areas are usually anemic and often develop severe anemia. This results from both shortened survival of uninfected RBCs and marked dyserythropoiesis. Anemia is a common consequence of antimalarial drug resistance, which results in repeated or continued infection.

Slight coagulation abnormalities are common in falciparum malaria, and mild thrombocytopenia is usual (a normal platelet count should question the diagnosis of malaria). Fewer than 5% of patients with severe malaria have significant bleeding with evidence of disseminated intravascular coagulation. Hematemesis from stress ulceration or acute gastric erosions also may occur rarely.

Liver Dysfunction Mild hemolytic jaundice is common in malaria. Severe jaundice is associated with *P. falciparum* infections; is more common among adults than among children; and results from hemolysis, hepatocyte injury, and cholestasis. Liver failure does not occur. When accompanied by other vital-organ dysfunction (often renal impairment), liver dysfunction carries a poor prognosis. Hepatic dysfunction contributes to hypoglycemia, lactic acidosis, and impaired drug metabolism. Occasional patients with falciparum malaria may develop deep jaundice (with hemolytic, hepatic, and cholestatic components) without evidence of other vital-organ dysfunction, in which case the prognosis is good.

Other Complications HIV/AIDS and malnutrition predispose to more severe malaria in nonimmune individuals. Malaria anemia is worsened by concurrent infections with intestinal helminths,

TABLE 224-4 Relative Incidence of Severe Complications of Falciparum Malaria

COMPLICATION	NONPREGNANT ADULTS	PREGNANT WOMEN	CHILDREN
Anemia	+	++	+++
Convulsions	+	+	+++
Hypoglycemia	+	+++	+++
Jaundice	+++	+++	+
Renal failure	+++	+++	—
Pulmonary edema	++	+++	+

Note: —, rare; +, infrequent; ++, frequent; +++, very frequent.

hookworm in particular. Approximately 6% of children diagnosed with severe malaria have concomitant bacteremia. In adults, the proportion is lower (<1%), except in those with very high parasite counts (>20% parasitemia). In areas of moderate and high malaria transmission, differentiating severe malaria from sepsis with incidental parasitemia in childhood is very difficult. In endemic areas, *Salmonella* spp. bacteremia has been associated specifically with *P. falciparum* infections. Chest infections and catheter-induced urinary tract infections are common among patients who are unconscious for >3 days. Aspiration pneumonia may follow generalized convulsions. The frequencies of complications of severe falciparum malaria are summarized in [Table 224-4](#).

■ MALARIA IN PREGNANCY

Malaria in early pregnancy causes fetal loss. In areas of high malaria transmission, falciparum malaria in primi- and secundigravid women is associated with low birth weight (average reduction, ~170 g) and consequently increased infant mortality rates. In general, infected mothers in areas of stable transmission remain asymptomatic despite intense accumulation of parasitized erythrocytes in the placental microcirculation. Maternal HIV infection predisposes pregnant women to more frequent and higher-density malaria infections, predisposes their newborns to congenital malarial infection, and exacerbates the reduction in birth weight associated with malaria.

In areas with unstable transmission of malaria, pregnant women are prone to severe infections and are particularly likely to develop high *P. falciparum* parasitemias complicated by anemia, hypoglycemia, and acute pulmonary edema. Fetal distress, premature labor, and stillbirth or low birth weight are common results. Fetal death is usual in severe malaria. Congenital malaria occurs in <5% of newborns of infected mothers; its frequency and the level of parasitemia are related directly to the timing of maternal infection and the parasite density in maternal blood and in the placenta. *P. vivax* malaria in pregnancy is also associated with a reduction in birth weight (average, 110 g), but in contrast to falciparum malaria, this effect is more pronounced in multigravid than in primigravid women. About 300,000 women die in childbirth yearly, with most deaths occurring in low-income countries; maternal death from hemorrhage at childbirth is correlated with malaria-induced anemia.

■ MALARIA IN CHILDREN

Most of the >400,000 deaths from falciparum malaria each year are in young African children. Convulsions, coma, hypoglycemia, metabolic acidosis, and severe anemia are relatively common among children with severe malaria, whereas deep jaundice, oliguric acute kidney injury, and acute pulmonary edema are unusual. Severely anemic children may present with labored deep breathing, which in the past has been attributed incorrectly to “anemic congestive cardiac failure” but in fact is usually caused by metabolic acidosis, sometimes compounded by hypovolemia. In general, children tolerate antimalarial drugs well and respond rapidly to treatment.

■ TRANSFUSION MALARIA

Malaria can be transmitted by blood transfusion, needlestick injury, or organ transplantation. The incubation period in these settings is often

short because there is no preerythrocytic stage of development, and thus there are no relapses of *P. vivax* and *P. ovale* infections. The clinical features and management of these cases are the same as for naturally acquired infections although primaquine is not needed for vivax or ovale malaria as there are no liver stages.

CHRONIC COMPLICATIONS OF MALARIA

■ HYPERREACTIVE MALARIAL SPLENOMEGALY

Chronic or repeated malarial infections produce hypergammaglobulinemia; normochromic, normocytic anemia; and, in certain situations, splenomegaly. Some residents of malaria-endemic areas in tropical countries exhibit an abnormal immunologic response to repeated infections that is characterized by massive splenomegaly, hepatomegaly, marked elevations in serum IgM and malarial antibody titers, hepatic sinusoidal lymphocytosis, and (in Africa) peripheral B-cell lymphocytosis. This syndrome has been associated with the production of cytotoxic IgM antibodies to CD8+ T lymphocytes, antibodies to CD5+ T lymphocytes, and an increase in the ratio of CD4+ to CD8+ T cells. These events may lead to uninhibited B cell production of IgM and the formation of cryoglobulins (IgM aggregates and immune complexes). This immunologic process stimulates lymphoid hyperplasia and clearance activity and eventually produces splenomegaly. Patients with hyperreactive malarial splenomegaly present with an abdominal mass or a dragging sensation in the abdomen and occasional sharp abdominal pains suggesting perisplenitis. There is usually anemia and some degree of pancytopenia (hypersplenism). In some cases, malaria parasites cannot be found in peripheral-blood smears by microscopy. Respiratory and skin infections are common and many patients die of overwhelming sepsis. Persons with hyperreactive malarial splenomegaly living in endemic areas should receive antimalarial chemoprophylaxis; the results are usually good. In nonendemic areas, antimalarial treatment is advised. Some cases have been mistaken for hematologic malignancy. However, in other cases refractory to therapy, clonal lymphoproliferation may develop and this can evolve into a malignant lymphoproliferative disorder.

■ QUARTAN MALARIAL NEPHROPATHY

Chronic or repeated infections with *P. malariae* (and possibly with other malarial species) may cause soluble immune complex injury to the renal glomeruli, resulting in the nephrotic syndrome. Other unidentified factors must contribute to this process since only a very small proportion of infected patients develop renal disease. The histologic appearance is that of focal or segmental glomerulonephritis with splitting of the capillary basement membrane. Subendothelial dense deposits are seen on electron microscopy, and immunofluorescence reveals deposits of complement and immunoglobulins and *P. malariae* antigens are often visible. A coarse-granular pattern of basement membrane immunofluorescent deposits (predominantly IgG3) with selective proteinuria carries a better prognosis than a fine-granular, predominantly IgG2 pattern with nonselective proteinuria. Quartan nephropathy is rarely reported nowadays. It usually responds poorly to treatment with either antimalarial agents or glucocorticoids and cytotoxic drugs.

■ BURKITT'S LYMPHOMA AND EPSTEIN-BARR VIRUS INFECTION

It is possible that malaria-related immune dysregulation provokes infection with lymphoma viruses. Childhood Burkitt's lymphoma is strongly associated with Epstein-Barr virus (EBV) and with high transmission of *P. falciparum*. Chronic *P. falciparum* malaria drives large numbers of EBV-infected cells through the lymph node germinal centers and deregulates activation-induced cytidine deaminase, resulting in DNA damage, c-myc translocations, and sometimes lymphoma.

DIAGNOSIS OF MALARIA

When a patient in or from a malarious area presents with fever, thick and thin blood smears should be prepared and examined immediately to confirm the diagnosis and identify the species of infecting parasite

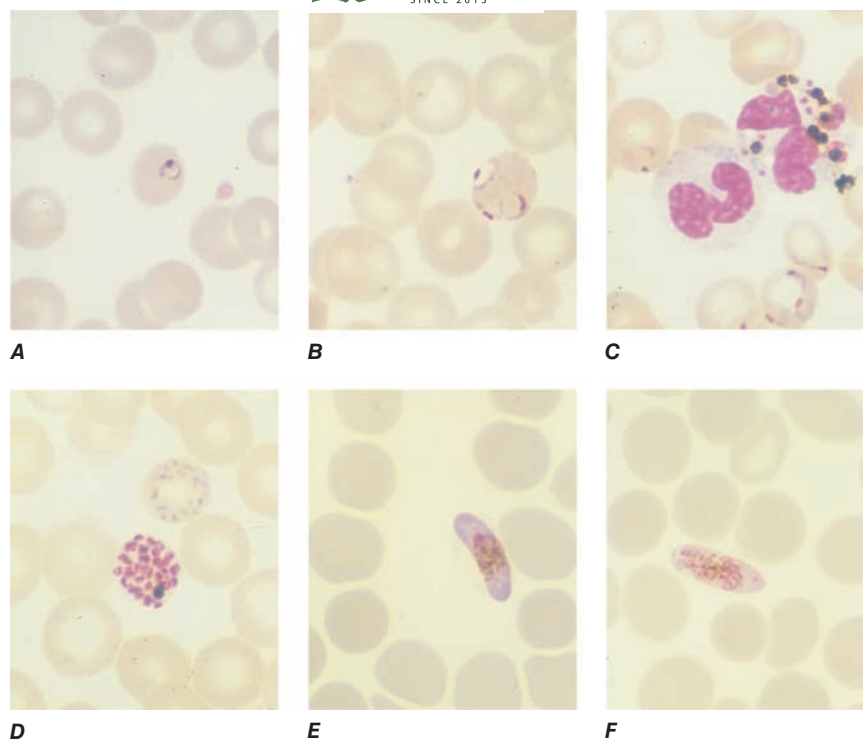


FIGURE 224-4 Thin blood films of *Plasmodium falciparum*. **A.** Young trophozoite. **B.** Old trophozoite. **C.** Trophozoites in erythrocytes and pigment in polymorphonuclear cells. **D.** Mature schizont. **E.** Female gametocyte. **F.** Male gametocyte. (Reproduced from *Bench Aids for the Diagnosis of Malaria Infections*, 2nd ed, with the permission of the World Health Organization.)

(Figs. 224-4 through 224-9). In general, if the blood smear is negative when examined by an experienced microscopist, the patient does not have malaria. If reliable microscopy is not available, a rapid test should be performed. Malaria is not a clinical diagnosis.

■ DEMONSTRATION OF THE PARASITE

The definitive diagnosis of malaria rests on the demonstration of asexual forms of the parasite in stained peripheral-blood smears.

Of the Romanowsky stains, Giemsa at pH 7.2 is preferred; Field's, Wright's, or Leishman's stain can also be used. Staining of parasites with the fluorescent dye acridine orange allows more rapid diagnosis of malaria (but not speciation of the infection) in patients with low-level parasitemia.

Both thin (Figs. 224-4 and 224-5) and thick (Figs. 224-6, 224-7, 224-8, and 224-9) blood smears should be examined. The thin blood smear should be air-dried, fixed in anhydrous methanol, then

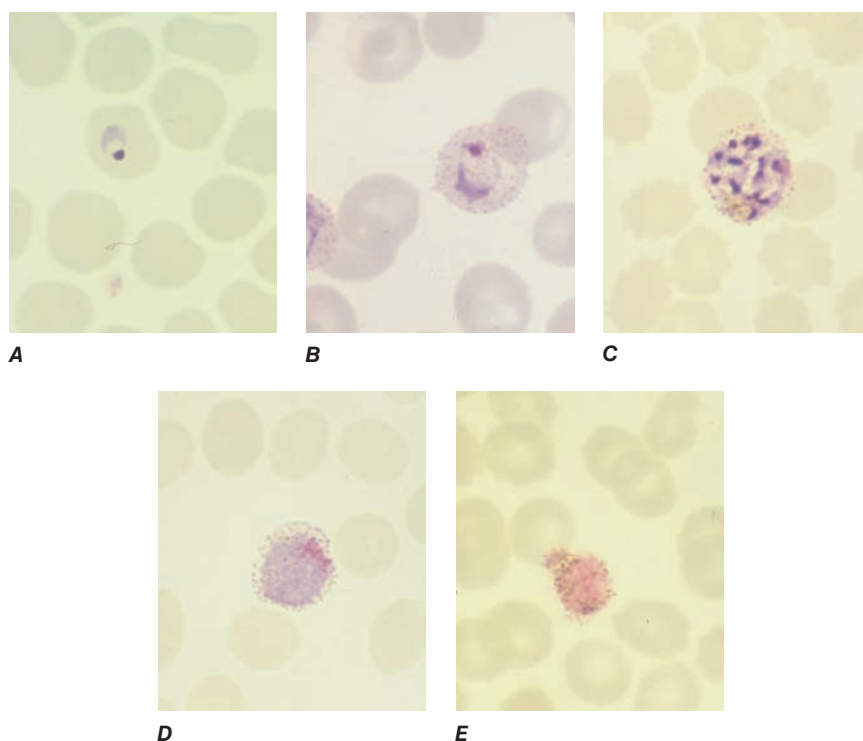


FIGURE 224-5 Thin blood films of *Plasmodium vivax*. **A.** Young trophozoite. **B.** Old trophozoite. **C.** Mature schizont. **D.** Female gametocyte. **E.** Male gametocyte. (Reproduced from *Bench Aids for the Diagnosis of Malaria Infections*, 2nd ed, with the permission of the World Health Organization.)

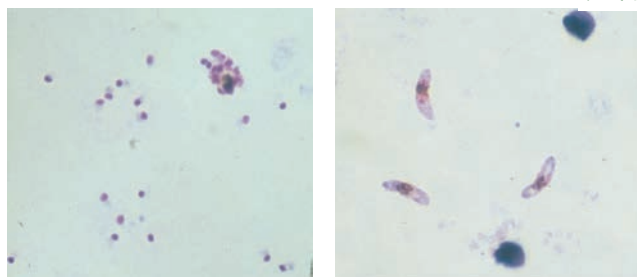
**A****B**

FIGURE 224-6 Thick blood films of *Plasmodium falciparum*. **A.** Trophozoites. **B.** Gametocytes. (Reproduced from *Bench Aids for the Diagnosis of Malaria Infections*, 2nd ed, with the permission of the World Health Organization.)

stained; the RBCs in the tail of the film should then be examined under oil immersion ($\times 1000$ magnification). The density of parasitemia is expressed as the number of parasitized erythrocytes per 1000 RBCs. The thick blood film should be of uneven thickness. The smear should be dried thoroughly and stained without fixing. As many layers of erythrocytes overlies one another and are lysed during the staining procedure, the thick film has the advantage of concentrating the parasites (by 40- to 100-fold compared with a thin blood film) and thus increasing diagnostic sensitivity. Both parasites and white blood cells (WBCs) are counted, and the number of parasites per unit volume is calculated from the total leukocyte count. Alternatively, a WBC count of 8000/ μL is assumed. This figure is converted to the number of parasitized erythrocytes per microliter. A minimum of 200 WBCs should be counted under oil immersion. Interpretation of blood smears, particularly thick films, requires some experience because artifacts are common. Before a thick smear is judged to be negative, 100–200 fields should be examined. In high-transmission areas, the presence of up to 10,000 parasites/ μL of blood may be tolerated without symptoms or signs in partially immune individuals. Thus, in these areas, the detection of low-density malaria parasitemia is sensitive but has low specificity in identifying malaria as the cause of illness. Because the prevalence of asymptomatic parasitemia is often high, low-density parasitemia is a common incidental finding in other conditions causing fever.

Rapid, simple, sensitive, and specific antibody-based diagnostic stick or card tests that detect *P. falciparum*-specific, histidine-rich protein 2 (PfHRP2), lactate dehydrogenase, or aldolase antigens in finger-prick blood samples are now being used widely in control programs (Table 224-5). Some of these rapid diagnostic tests (RDTs) carry a second antibody (either pan-malaria or *P. vivax*-specific) and so distinguish falciparum malaria from the less dangerous malarias. PfHRP2-based RDTs may remain positive for several weeks after acute infection. This prolonged positivity is a disadvantage in high-transmission areas where infections are frequent but helps in the diagnosis of severe malaria in patients who have taken antimalarial drugs and cleared peripheral parasitemia but who still have a strongly positive PfHRP2 test. A disadvantage of RDTs is that they do not quantify parasitemia. Widespread use of PfHRP2 RDTs has put strong selection

pressure on *P. falciparum* populations in some areas, leading to an increased prevalence of mutant parasites that are not detected by the current generation of PfHRP2-based tests.

The relationship between parasite density and prognosis is complex and variable; in general, patients with $>10^5$ parasites/ μL are at increased risk of dying, but nonimmune patients may die with much lower counts, and partially immune persons may tolerate parasitemia levels many times higher with only minor symptoms. In severe malaria, a poor prognosis is indicated by a predominance of more mature *P. falciparum* parasites (i.e., $>20\%$ of parasites with visible pigment) in the peripheral-blood film or by the presence of phagocytosed malarial pigment in $>5\%$ of neutrophils (an indicator of recent schizogony). In *P. falciparum* infections, gametocytemia peaks 1 week after the peak of asexual parasite densities. Because the mature gametocytes of *P. falciparum* (unlike those of other plasmodia) are not affected by most anti-malarial drugs, their persistence does not mean there is drug resistance or a need to re-treat if a full course of appropriate treatment has been given. Phagocytosed malarial pigment seen inside peripheral-blood monocytes may provide a clue to recent infection if malaria parasites are not detectable. After parasite clearance, this intraphagocytic malarial pigment is often evident for several days in peripheral-blood films or for longer in bone marrow aspirates or smears of fluid expressed after intradermal puncture.

Molecular diagnosis by polymerase chain reaction (PCR) amplification of parasite nucleic acid is more sensitive than microscopy or rapid diagnostic tests for detecting malaria parasites and defining malarial species. While currently impractical in the standard clinical setting, PCR is used in reference centers in endemic areas. In epidemiologic surveys, ultrasensitive PCR detection has proved very useful in identifying asymptomatic infections as control and eradication programs drive parasite prevalences down to very low levels. Serologic diagnosis with either indirect fluorescent antibody or enzyme-linked immunosorbent assays is useful for screening of prospective blood donors and may prove useful as a measure of transmission intensity in future epidemiologic studies. Serology has no place in the diagnosis of acute illness.

■ LABORATORY FINDINGS IN ACUTE MALARIA

Normochromic, normocytic anemia is usual. The leukocyte count is generally normal, although it may be raised in very severe infections. There is slight monocytosis, lymphopenia, and eosinopenia, with reactive lymphocytosis and eosinophilia in the weeks after acute infection. The platelet count is usually reduced to $\sim 10^5/\mu\text{L}$. The erythrocyte sedimentation rate, plasma viscosity, and levels of C-reactive protein and other acute-phase proteins are elevated. Severe infections may be accompanied by prolonged prothrombin and partial thromboplastin times and by more severe thrombocytopenia. Antithrombin III levels are reduced even in mild infection. In uncomplicated malaria, plasma concentrations of electrolytes, blood urea nitrogen (BUN), and creatinine are usually normal. Findings in severe malaria may include metabolic acidosis, with low plasma concentrations of glucose, sodium, bicarbonate, phosphate, and albumin, together with elevations in lactate, BUN, creatinine, urate, muscle and liver enzymes, and conjugated and unconjugated bilirubin. Hypergammaglobulinemia is usual in

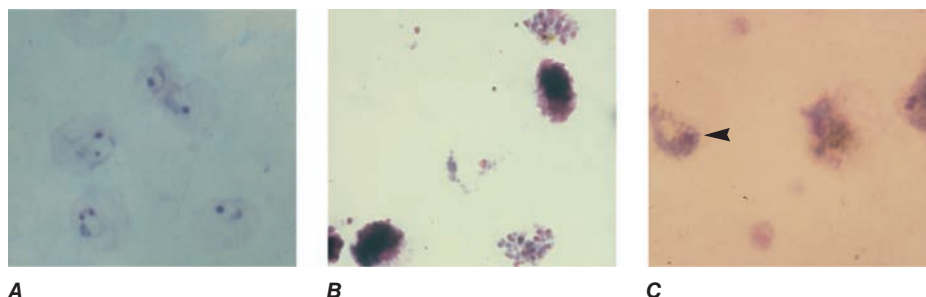
**A****B****C**

FIGURE 224-7 Thick blood films of *Plasmodium vivax*. **A.** Trophozoites. **B.** Schizonts. **C.** Gametocytes. (Reproduced from *Bench Aids for the Diagnosis of Malaria Infections*, 2nd ed, with the permission of the World Health Organization.)

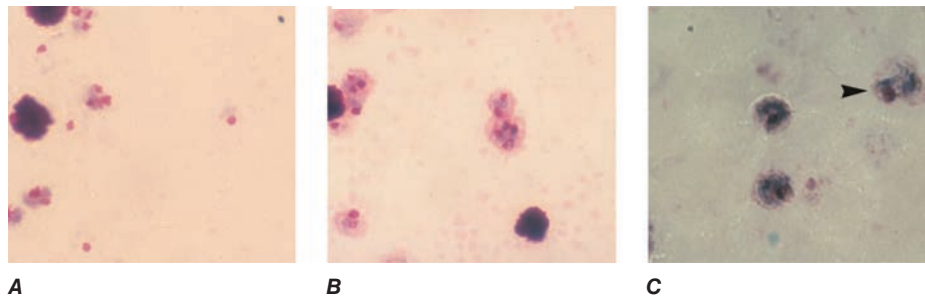


FIGURE 224-8 Thick blood films of *Plasmodium ovale*. **A.** Trophozoites. **B.** Schizonts. **C.** Gametocytes. (Reproduced from *Bench Aids for the Diagnosis of Malaria Infections*, 2nd ed, with the permission of the World Health Organization.)

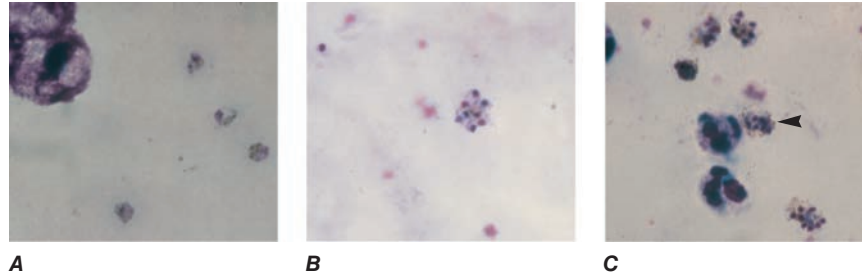


FIGURE 224-9 Thick blood films of *Plasmodium malariae*. **A.** Trophozoites. **B.** Schizonts. **C.** Gametocytes. (Reproduced from *Bench Aids for the Diagnosis of Malaria Infections*, 2nd ed, with the permission of the World Health Organization.)

TABLE 224-5 Standard Methods for the Diagnosis of Malaria ^a			
METHOD	PROCEDURE	ADVANTAGES	DISADVANTAGES
Thick blood film ^b	Blood should be uneven in thickness but thin enough that the hands of a watch can be read through part of the spot. Stain dried, unfixed blood spot with Giemsa, Field's, or another Romanowsky stain. Count number of asexual parasites per 200 WBCs (or per 500 WBC at low densities). Count and report gametocytes separately. ^c	Sensitive (0.001% parasitemia); species specific; inexpensive	Requires experience (artifacts may be misinterpreted as low-level parasitemia); underestimates true count
Thin blood film ^d	Stain fixed smear with Giemsa, Field's, or another Romanowsky stain. Count number of RBCs containing asexual parasites per 1000 RBCs. In severe malaria, assess stage of parasite development and count neutrophils containing malaria pigment. ^e Count and report gametocytes separately. ^c	Rapid; species specific; inexpensive; in severe malaria, provides prognostic information ^e	Insensitive (<0.05% parasitemia); uneven distribution of <i>P. vivax</i> , as enlarged infected red cells concentrate at leading edge
PfHRP2 dipstick or card test	A drop of blood is placed on the stick or card, which is then immersed in washing solutions. Monoclonal antibody capture of parasitic antigens reads out as a colored band.	Robust and relatively inexpensive; rapid; sensitivity similar to or slightly lower than that of thick films (~0.001% parasitemia)	Detects only <i>Plasmodium falciparum</i> ; remains positive for weeks after high-density infections ^f ; does not quantitate <i>P. falciparum</i> parasitemia; evasion of detection by certain strains due to polymorphisms in <i>HRP2</i> gene
<i>Plasmodium</i> LDH dipstick or card test	A drop of blood is placed on the stick or card, which is then immersed in washing solutions. Monoclonal antibody capture of parasitic antigens reads out as two colored bands. One band is genus specific (all malarias), or <i>P. vivax</i> specific, and the other band is specific for <i>P. falciparum</i> .	Rapid; sensitivity similar to or slightly lower than that of thick films for <i>P. falciparum</i> (~0.001% parasitemia)	May miss low-level parasitemia with <i>P. vivax</i> , <i>P. ovale</i> , and <i>P. malariae</i> and may not speciate these organisms; does not quantitate <i>P. falciparum</i> parasitemia; lower sensitivity for detection of <i>P. knowlesi</i> , which may be misidentified as <i>P. falciparum</i>
Microtube concentration methods with acridine orange staining	Blood is collected in a specialized tube containing acridine orange, anticoagulant, and a float. After centrifugation, which concentrates the parasitized cells around the float, fluorescence microscopy is performed.	Sensitivity similar or superior to that of thick films (~0.001% parasitemia); ideal for processing large numbers of samples rapidly	Does not speciate or quantitate; requires fluorescence microscopy

^aMalaria cannot be diagnosed clinically with accuracy, but treatment should be started on clinical grounds if laboratory confirmation is likely to be delayed. In areas of the world where malaria is endemic and transmission rates are high, low-level asymptomatic parasitemia is common in otherwise healthy people. Thus, malaria may not be the cause of a fever, although in this context, the presence of >10,000 parasites/μL (~0.2% parasitemia) does indicate that malaria is likely to be the cause. Antibody and polymerase chain reaction (PCR) tests have no role in the diagnosis of malaria except that PCR is increasingly used for genotyping and speciation in mixed infections and for detection of low-level parasitemia in asymptomatic residents of endemic areas. ^bAsexual parasites/200 WBCs × 40 = parasite count/μL (assumes a WBC count of 8000/μL). See Figs. 224-6 through 224-9. ^c*P. falciparum* gametocytemia may persist for days or weeks after clearance of asexual parasites. Gametocytemia without asexual parasitemia does not indicate active infection. ^dParasitized RBCs (/1000) × hematocrit × 125.6 = parasite count/μL. See Figs. 224-4 and 224-5. ^eThe presence of >100,000 parasites/μL (~2% parasitemia) is associated with an increased risk of severe malaria, but some patients have severe malaria with lower counts. At any level of parasitemia, the finding that >50% of parasites are tiny rings (cytoplasm thickness less than half of nucleus width) carries a relatively good prognosis. In a severely ill patient, the presence of visible pigment in >20% of parasites or of phagocytosed pigment in >5% of polymorphonuclear leukocytes (indicating massive recent schizogony) carries a worse prognosis. ^fPersistence of PfHRP2 is a disadvantage in high-transmission settings, where many asymptomatic people have positive tests, but can be used to diagnostic advantage in low-transmission settings when a sick patient has previously received unknown treatment (which, in endemic areas, often consists of antimalarial drugs). In this situation, a positive PfHRP2 test indicates that the illness is falciparum malaria, even if the blood smear is negative.

Abbreviations: LDH, lactate dehydrogenase; PfHRP2, *P. falciparum* histidine-rich protein 2; RBCs, red blood cells; WBCs, white blood cells.

1730 immune and semi-immune subjects living in malaria-endemic areas. Urinalysis generally gives normal results. In adults and children with cerebral malaria, the mean cerebrospinal fluid (CSF) opening pressure at lumbar puncture is ~160 mm H₂O; usually the CSF content is normal or there is a slight elevation of total protein level (<1.0 g/L [<100 mg/dL]) and cell count (<20/ μ L).

TREATMENT

Malaria

Patients with severe malaria and those unable to take oral drugs should receive parenteral antimalarial therapy immediately (Table 224-6). Antimalarial drug susceptibility testing can be performed but is rarely available, has poor predictive value in an individual case, and yields results too slowly to influence the choice of treatment. If there is any doubt about the resistance status of the infecting organism, it should be considered resistant.

The World Health Organization (WHO) recommends artemisinin-based combination therapy (ACT) as first-line treatment for uncomplicated *P. falciparum* malaria in malaria-endemic areas. ACT is also the recommended first-line treatment for *P. knowlesi* infections, and either chloroquine or an ACT is recommended for the other malarias. The choice of ACT partner drug depends on the likely sensitivity of the infecting parasites. Artemisinin-based combinations are sometimes unavailable in temperate countries, where treatment recommendations are limited to the registered available drugs. Despite increasing evidence of chloroquine resistance in *P. vivax* (from parts of Indonesia, Oceania, eastern and southern Asia, and Central and South America), chloroquine remains an effective treatment for *P. vivax* malaria in many areas and for *P. ovale* and *P. malariae* infections everywhere.

Artemisinin resistance in *P. falciparum* has emerged in Southeast Asia over the past decade and has been followed by piperazine and mefloquine resistance. ACTs are failing in Cambodia, Vietnam, and the border regions of Thailand. Significant artemisinin resistance is

TABLE 224-6 Regimens for the Treatment of Malaria^a

TYPE OF DISEASE OR TREATMENT	REGIMEN(S)
Uncomplicated Malaria	
Known chloroquine-sensitive strains of <i>Plasmodium vivax</i> , <i>P. malariae</i> , <i>P. ovale</i> , <i>P. falciparum</i> ^b	Chloroquine (10 mg of base/kg stat followed by 5 mg/kg at 12, 24, and 36 h or by 10 mg/kg at 24 h and 5 mg/kg at 48 h) or Amodiaquine (10–12 mg of base/kg qd for 3 days)
Radical treatment for <i>P. vivax</i> or <i>P. ovale</i> infection (prevention of relapse)	In addition to chloroquine or amodiaquine or ACT, primaquine (0.5 mg of base/kg qd in Southeast Asia and Oceania [total dose 7 mg/kg] and 0.25 mg/kg elsewhere [total dose 3.5 mg/kg]) should be given for 14 days to prevent relapse. ^c In mild G6PD deficiency, 0.75 mg of base/kg should be given once weekly for 8 weeks. Primaquine should not be given in severe G6PD deficiency.
<i>P. falciparum</i> malaria ^c	Artesunate ^{d,e} (4 mg/kg qd for 3 days) plus sulfadoxine (25 mg/kg)/pyrimethamine (1.25 mg/kg) as a single dose or Artesunate ^d (4 mg/kg qd for 3 days) plus amodiaquine (10 mg of base/kg qd for 3 days) ^{d,e} or Artemether-lumefantrine ^d (1.5/9 mg/kg bid for 3 days with food) or Artesunate ^d (4 mg/kg qd for 3 days) plus mefloquine (24–25 mg of base/kg—either 8 mg/kg qd for 3 days or 15 mg/kg on day 2 and then 10 mg/kg on day 3) ^f or DHA-piperaquine ^d (target dose: 4/24 mg/kg qd for 3 days in children weighing <25 kg and 4/18 mg/kg qd for 3 days in persons weighing ≥25 kg) or Artesunate-pyronaridine ^d (4/12 mg/kg qd for 3 days)
Second-line treatment/treatment of imported malaria	Artesunate ^e (2 mg/kg qd for 7 days) or quinine (10 mg of salt/kg tid for 7 days) plus 1 of the following 3: 1. Tetracycline ^g (4 mg/kg qid for 7 days) 2. Doxycycline ^g (3 mg/kg qd for 7 days) 3. Clindamycin (10 mg/kg bid for 7 days) or Atovaquone-proguanil (20/8 mg/kg qd for 3 days with food)
Severe Falciparum Malaria^{a,h}	
	Artesunate ^e (2.4 mg/kg stat IV followed by 2.4 mg/kg at 12 and 24 h and then daily if necessary; for children weighing <20 kg, give 3 mg/kg per dose) or, if unavailable, Artemether ^e (3.2 mg/kg stat IM followed by 1.6 mg/kg qd) or, if unavailable, Quinine dihydrochloride (20 mg of salt/kg infused over 4 h, followed by 10 mg of salt/kg infused over 2–8 h q8h ⁱ)

^aIn endemic areas where malaria transmission is low, except in pregnant women and infants, a single dose of primaquine (0.25 mg of base/kg) should be added as a gametocytocide to all falciparum malaria treatments to prevent transmission. This addition is considered safe, even in G6PD deficiency. ^bVery few areas now have chloroquine-sensitive *P. falciparum* malaria. ^cRecent large studies indicate that these total doses can be condensed into 7-day primaquine regimens. ^dIn areas where the partner drug to artesunate is known to be effective. Fixed-dose co-formulated combinations are available. The World Health Organization recommends artemisinin combination regimens as first-line therapy for falciparum malaria in all tropical countries and advocates use of fixed-dose combinations. ^eArtemisinin derivatives are not readily available in some temperate countries. ^fTetracycline and doxycycline should not be given to pregnant women or to children <8 years of age. ^gOral treatment should be substituted as soon as the patient recovers sufficiently to take fluids by mouth. ^hArtesunate is the drug of choice when available. The data from large studies in Southeast Asia showed a 35% lower mortality rate than with quinine, and very large studies in Africa showed a 22.5% reduction in mortality rate compared with quinine. The doses of artesunate in children weighing <20 kg should be 3 mg/kg. ⁱA loading dose should not be given if therapeutic doses of quinine have definitely been administered in the previous 24 h. ^jInfusions can be given in 0.9% saline and 5–10% dextrose in water. Infusion rates for quinine should be carefully controlled.

Abbreviations: ACT, artemisinin combination therapy; DHA, dihydroartemisinin; G6PD, glucose-6-phosphate dehydrogenase.

now prevalent throughout the Greater Mekong subregion and there is recent clear evidence for the emergence of artemisinin resistance in East Africa (Rwanda, Uganda). Falsified or substandard antimalarial drugs are sold in many Asian and African countries and may be the cause of treatment failures. Characteristics of antimalarial drugs are shown in [Table 224-7](#).

SEVERE MALARIA

In large randomized controlled clinical trials, parenteral artesunate, a water-soluble artemisinin derivative, has reduced severe falciparum malaria mortality rates by 35% in Asian adults and children and by 22.5% in African children compared with quinine treatment. Artesunate therefore is now the drug of choice for all patients with severe malaria everywhere. Artesunate is given by IV injection but is also absorbed rapidly following IM injection. Artemether and the closely related drug artemotil (arteether) are oil-based formulations given by IM injection; they are erratically absorbed and do not confer the same survival benefit as artesunate. A rectal formulation of artesunate has been developed as a community-based pre-referral treatment for patients in the rural tropics who cannot take oral medications. Pre-referral administration of rectal artesunate has been shown to decrease mortality rates among severely ill children without access to immediate parenteral treatment. IV artesunate has been approved by the U.S. Food and Drug Administration for emergency use in severe malaria and can be obtained through the Centers for Disease Control and Prevention (CDC) Drug Service (see end of chapter for contact information). The antiarrhythmic quinidine gluconate was used to treat severe malaria in the United States previously, but manufacturing was discontinued in 2019; artesunate is much more effective and safer. Although parenteral quinine is steadily being replaced by parenteral artesunate in endemic areas, it still has a role in the very few cases of artemisinin-resistant severe falciparum malaria from Southeast Asia, where both artesunate and quinine are given together in full doses.

Severe falciparum malaria constitutes a medical emergency requiring intensive nursing care and careful management. Frequent evaluation of the patient's condition is essential. Adjunctive treatments such as high-dose glucocorticoids, urea, heparin, dextran, desferrioxamine, antibody to tumor necrosis factor α , high-dose phenobarbital (20 mg/kg), mannitol, or large-volume fluid or albumin boluses have proved either ineffective or harmful in clinical trials and should not be used. In acute renal failure or severe metabolic acidosis, hemofiltration or hemodialysis should be started as early as possible.

In severe malaria, parenteral antimalarial treatment should be started immediately. Artesunate, given by either IV or IM injection, is simple to administer, very safe, and rapidly effective. It does not require dose adjustments in liver dysfunction or renal failure. It should be used in pregnant women with severe malaria. If artesunate is unavailable and artemether or quinine is used, an initial loading dose must be given so that therapeutic concentrations are reached as soon as possible. Quinine causes dangerous hypotension if injected rapidly and so must be administered carefully by rate-controlled infusion only. If this approach is not possible, quinine may be given by deep IM injections into the anterior thigh. The optimal therapeutic range for quinine in severe malaria is not known with certainty, but total plasma concentrations of 8–15 mg/L for quinine are effective and do not cause serious toxicity. The systemic clearance and apparent volume of distribution of quinine are markedly reduced and plasma protein binding is increased in severe malaria, so that the blood concentrations attained with a given dose are higher. If the patient remains seriously ill or in acute renal failure for >2 days, maintenance doses of quinine should be reduced by 30–50% to prevent toxic accumulation of the drug. The initial dose should never be reduced. Convulsions should be treated promptly with IV (or rectal) benzodiazepines. The role of prophylactic anti-convulsants in children is uncertain. If respiratory support is not available, a full loading dose of phenobarbital (20 mg/kg) to prevent convulsions should not be given as it may cause respiratory arrest.

When the patient is unconscious, the blood glucose level should be measured every 4–6 h. All patients should receive a continuous infusion of dextrose, and blood concentrations ideally should be maintained above 4 mmol/L. Hypoglycemia (<2.2 mmol/L or 40 mg/dL) should be treated immediately with bolus glucose. The parasite count and hematocrit should be measured every 6–12 h. Anemia develops rapidly. There is uncertainty as to the thresholds for transfusion as there is some evidence that moderate anemia may be beneficial in a patient with severe malaria and vital organ dysfunction. It has been recommended that if the hematocrit falls to <20%, whole blood (preferably fresh) or packed cells should be transfused slowly, with careful attention to circulatory status. In areas with higher malaria transmission, where blood for transfusion is in short supply, a threshold of 15% is widely used. Renal function should be checked at least daily. Children presenting with very severe anemia (hemoglobin <4 g/dL) and acidotic breathing require immediate blood transfusion. Accurate assessment is vital. Management of fluid balance is difficult in severe malaria, particularly in adults, because of the thin dividing line between overhydration (leading to pulmonary edema) and underhydration (contributing to renal impairment). Fluid balance management is different from that in sepsis: fluid boluses are potentially dangerous in severe malaria. Nasogastric feeding should be delayed in nonintubated patients (for 60 h in adults and 36 h in children) to reduce the risk of aspiration pneumonia. As soon as the patient can take fluids, oral therapy should be substituted for parenteral treatment and a full 3-day course of ACT given. Mefloquine should be avoided as follow-on treatment for severe malaria because of the increased risk of post-malaria neurologic syndrome.

In areas of high transmission of both *P. falciparum* and *P. vivax* (the island of New Guinea), severe and potentially life-threatening anemia is common among children, and both species contribute. Elsewhere, severe vivax malaria may occur but is uncommon. Many patients have had comorbidities contributing to vital-organ dysfunction.

P. knowlesi can cause severe disease associated with high parasite densities. Acute kidney injury, respiratory distress, and shock have all been described, but cerebral malaria does not occur. Treatment for severe vivax and knowlesi malaria should follow the recommendations given for falciparum malaria.

UNCOMPLICATED MALARIA

P. falciparum and *P. knowlesi* infections should be treated with an artemisinin-based combination because of their propensity for high parasite densities and severe disease. Infections with sensitive strains of *P. vivax*, *P. malariae*, and *P. ovale* should be treated either with an ACT or oral chloroquine (total dose, 25 mg of base/kg). The ACT regimens now recommended are safe and effective in adults, children, and pregnant women. The rapidly eliminated artemisinin component is usually an artemisinin derivative (artesunate, artemether, or dihydroartemisinin) given for 3 days, and the partner drug is usually a more slowly eliminated antimalarial to which *P. falciparum* in the area is sensitive. Six ACT regimens are currently recommended by the WHO: artemether-lumefantrine, artesunate-mefloquine, dihydroartemisinin-piperaquine, artesunate-sulfadoxine-pyrimethamine, artesunate-amodiaquine, and artesunate-pyronaridine. In areas of low malaria transmission, a single dose of primaquine (0.25 mg/kg) should be added to ACT as a *P. falciparum* gametocytocide to reduce the transmissibility of the infection. This low dose of primaquine is safe even in G6PD deficiency. Pregnant women should not be given primaquine. Atovaquone-proguanil is highly effective everywhere, although it is seldom used in endemic areas because of its high cost and the propensity for rapid emergence of resistance. Recovery is slower after atovaquone-proguanil treatment than after ACT. Of great concern is the spread of artemisinin-resistant *P. falciparum* in the Greater Mekong subregion of Southeast Asia. Infections with these resistant parasites are cleared slowly from the blood, with parasite clearance half-lives over 5 hours and clearance times typically exceeding 3 days. Cure rates with ACT have fallen to unacceptably

TABLE 224-7 Properties of Antimalarial Drugs

DRUG(S) ^a	PHARMACOKINETIC PROPERTIES	ANTIMALARIAL ACTIVITY	MINOR TOXICITY	MAJOR TOXICITY
Quinine	Good oral and IM absorption (quinine); <i>Cl</i> and <i>V_d</i> reduced, but plasma protein binding (principally to α 1 acid glycoprotein) increased (90%) in malaria; quinine $t_{1/2}$: 16 h in malaria, 11 h in healthy persons	Acts mainly on trophozoite blood stage; kills gametocytes of <i>P. vivax</i> , <i>P. ovale</i> , and <i>P. malariae</i> (but not <i>P. falciparum</i>); no action on liver stages	<i>Common</i> : cinchonism (tinnitus, high-tone hearing loss, nausea, vomiting, dysphoria, postural hypotension); ECG QT interval prolongation (usually by <10%). <i>Rare</i> : diarrhea, visual disturbance, rashes. <i>Note</i> : very bitter taste	<i>Common</i> : hypoglycemia. <i>Rare</i> : hypotension, blindness, deafness, cardiac arrhythmias, thrombocytopenia, hemolysis, hemolytic-uremic syndrome, vasculitis, cholestatic hepatitis, neuromuscular paralysis.
Chloroquine	Good oral absorption, very rapid IM and SC absorption; complex pharmacokinetics; enormous <i>Cl</i> and <i>V_d</i> (unaffected by malaria); blood concentration profile determined by distribution processes in malaria; $t_{1/2}$: 1–2 months. Active desethyl metabolite about 25% of parent drug concentrations	As for quinine, but acts slightly earlier in asexual cycle	<i>Common</i> : nausea, dysphoria, pruritus in dark-skinned patients, postural hypotension, ECG QT prolongation. <i>Rare</i> : accommodation difficulties, keratopathy, hypoglycemia, rash. <i>Note</i> : bitter taste but usually well tolerated	<i>Acute</i> : hypotensive shock (parenteral), cardiac arrhythmias, neuropsychiatric reactions. <i>Chronic</i> : retinopathy (cumulative dose, >100 g), skeletal and cardiac myopathy
Piperaquine	Adequate oral absorption, may be enhanced by fats; similar pharmacokinetics to chloroquine; $t_{1/2}$: 21–28 days	As for chloroquine; retains activity against multidrug-resistant <i>P. falciparum</i> , but resistance has emerged in Southeast Asia	Occasional epigastric pain, diarrhea, ECG QT prolongation	None identified
Amodiaquine	Good oral absorption; largely converted to active metabolite desethylamodiaquine; $t_{1/2}$: 4–5 days	As for chloroquine, but more active against chloroquine-resistant <i>P. falciparum</i>	Nausea (tastes better than chloroquine), dysphoria, headache, bradycardia, ECG QT prolongation	Agranulocytosis; hepatitis, mainly with prophylactic use; should not be used with efavirenz
Primaquine	Complete oral absorption; active metabolite produced mainly via CYP2D6; $t_{1/2}$: 5–7 h	Radical cure; eradicates hepatic forms of <i>P. vivax</i> and <i>P. ovale</i> ; kills <i>P. falciparum</i> gametocytes development; kills developing liver stages of all species	Nausea, vomiting, diarrhea, abdominal pain, hemolysis, methemoglobinemia	Serious hemolytic anemia in severe G6PD deficiency; hemoglobinuria
Mefloquine	Adequate oral absorption; no parenteral preparation; $t_{1/2}$: 14–20 days (shorter in malaria)	As for quinine	Nausea, giddiness, dysphoria, fuzzy thinking, sleeplessness, nightmares, sense of dissociation	Neuropsychiatric reactions, convulsions, encephalopathy
Lumefantrine	Highly variable absorption related to fat intake; $t_{1/2}$: 3–4 days	As for quinine	None identified	None identified
Artemisinin and derivatives (artemether, artesunate)	Good oral absorption; good absorption of IM artesunate but slow and variable absorption of IM artemether; artesunate and artemether biotransformed to active metabolite dihydroartemisinin; all drugs eliminated very rapidly; $t_{1/2}$: <1 h	Broader stage specificity and more rapid than other drugs; no action on liver stages; kills all but fully mature gametocytes of <i>P. falciparum</i>	Reduction in reticulocyte count (but not anemia); neutropenia at high doses; in some cases, delayed anemia after treatment of severe malaria with hyperparasitemia	Anaphylaxis, urticaria, fever
Pyrimethamine	Good oral absorption, variable IM absorption; $t_{1/2}$: 4 days	For blood stages, acts mainly on mature forms; causal prophylactic	Well tolerated	Megaloblastic anemia, pancytopenia, pulmonary infiltration
Proguanil ^b (chloroguanide)	Good oral absorption; biotransformed to active metabolite cycloguanil; $t_{1/2}$: 16 h; biotransformation reduced by oral contraceptive use and in pregnancy	Causal prophylactic; not used alone for treatment	Well tolerated; mouth ulcers and rare alopecia	Megaloblastic anemia in renal failure
Atovaquone ^b	Highly variable absorption related to fat intake; $t_{1/2}$: 30–70 h	Acts mainly on trophozoite blood stage	None identified	None identified
Tetracycline, doxycycline ^c	Excellent absorption; $t_{1/2}$: 8 h for tetracycline, 18 h for doxycycline	Weak antimalarial activity; should not be used alone for treatment	Gastrointestinal intolerance, deposition in growing bones and teeth (tetracycline), photosensitivity, moniliasis, benign intracranial hypertension	Renal failure in patients with impaired renal function (tetracycline)
Pyronaridine	Rapid variable absorption, large <i>V_d</i> ; $t_{1/2}$: 12–14 days	Acts mainly on trophozoite blood stage; kills gametocytes of <i>P. vivax</i> , <i>P. ovale</i> , and <i>P. malariae</i> (but not <i>P. falciparum</i>); no action on liver stages	Gastrointestinal intolerance, anemia, transient elevation of aminotransferases, hypoglycemia, headache	None identified
Arterolane	$t_{1/2}$: 3 h	Broad stage specificity; no action on liver stages; kills all but fully mature gametocytes of <i>P. falciparum</i>	Gastrointestinal intolerance, transient elevation of aminotransferases	None identified

^aSeveral antimalarial drugs are formulated as different salts (e.g., phosphate, sulfate, hydrochloride) and are therefore prescribed as base equivalents. For example, chloroquine phosphate 250 salt contains 155 mg base equivalent. It is very important to check when prescribing that the correct dose is being given. ^bAtovaquone and proguanil are prescribed as a fixed-dose combination. This and proguanil alone should not be given if the estimated glomerular filtration rate is <30 mL/min. ^cTetracycline and doxycycline should not be given to pregnant women or to children <8 years of age.

Abbreviations: *Cl*, systemic clearance; ECG, electrocardiogram; G6PD, glucose-6-phosphate dehydrogenase; *V_d*, total apparent volume of distribution.

low levels in some areas. Triple antimalarial combinations are under evaluation with promising results to date.

The 3-day ACT regimens are all well tolerated, although mefloquine is associated with increased rates of vomiting and dizziness. As second-line treatments for recrudescence following first-line therapy, a different ACT regimen may be given; another alternative is a 7-day course of either artesunate or quinine plus tetracycline, doxycycline, or clindamycin. Tetracycline and doxycycline are not recommended to treat pregnant women or children <8 years of age, however evidence for doxycycline toxicity in these groups is weak. Oral quinine is extremely bitter and regularly produces cinchonism comprising tinnitus, high-tone deafness, nausea, vomiting, and dysphoria. Clinical responses are slower than those following ACT. Adherence is poor with the required 7-day regimens of quinine.

Patients should be monitored for vomiting for 1 h after the administration of any oral antimalarial drug. If there is vomiting, the dose should be repeated. Symptom-based treatment, with acetaminophen (paracetamol) administration, lowers fever and thereby reduces the patient's propensity to vomit these drugs. Minor central nervous system reactions (nausea, dizziness, sleep disturbances) are common. The incidence of serious adverse neuropsychiatric reactions to mefloquine treatment is ~1 in 1000 in Asia but may be as high as 1 in 200 among Africans and white ethnic groups. All the antimalarial quinolines (chloroquine, amodiaquine, mefloquine, and quinine) exacerbate the orthostatic hypotension associated with malaria, and all are tolerated better by children than by adults. Pregnant women, young children, patients unable to tolerate oral therapy, and nonimmune individuals (e.g., travelers) with suspected malaria should be evaluated carefully and hospitalization considered. If there is any doubt as to the identity of the infecting malarial species, treatment for falciparum malaria should be given. A negative blood smear read by an experienced microscopist makes malaria very unlikely but does not rule it out completely; thick blood films should be checked again 1 and 2 days later to exclude the diagnosis. Nonimmune patients receiving treatment for malaria should have daily parasite counts performed until the thick films are negative for asexual parasite stages. If the level of parasitemia does not fall below 25% of the admission value in 72 h or if parasitemia has not cleared by 7 days (and adherence is assured), drug resistance is likely and the regimen should be changed.

To eradicate persistent liver stages and prevent relapse (radical treatment), primaquine (0.5 mg of base/kg in East Asia and Oceania and 0.25 mg/kg elsewhere) should be given once daily for 14 days to patients with *P. vivax* or *P. ovale* infection after laboratory tests for G6PD deficiency have proved negative. The same total dose may be given over 7 days. If the patient has a mild variant of G6PD deficiency, primaquine can be given in a dose of 0.75 mg of base/kg (maximum, 45 mg) once weekly for 8 weeks. Pregnant women with vivax or ovale malaria should not be given primaquine but should receive suppressive prophylaxis with chloroquine (5 mg of base/kg per week) until delivery, after which radical treatment can be given. The slowly eliminated 8-aminoquinoline tafenoquine has been registered in some countries. This allows radical cure to be given in a single dose. The consequent risk of protracted hemolysis in G6PD deficiency, including in female heterozygotes who may test as normal with current G6PD screens (which detect <30–40% of normal enzyme activity), requires that all patients should have a quantitative test of G6PD activity before receiving tafenoquine. Only those with >70% of normal activity should receive the drug. Radical curative efficacy is lower than with primaquine in Southeast Asia.

MANAGEMENT OF COMPLICATIONS

Acute Renal Failure If plasma levels of BUN or creatinine rise despite adequate rehydration, fluid administration should be restricted to prevent volume overload. As in other forms of hypercatabolic acute renal failure, renal replacement therapy is best performed early (**Chap. 310**). Hemofiltration and hemodialysis are more effective than peritoneal dialysis and are associated with lower mortality risk. Some patients with renal impairment pass small

volumes of urine sufficient to allow control of fluid balance; these cases can be managed conservatively if other indications for dialysis do not arise. Renal function usually improves within days, but full recovery may take weeks.

Acute Pulmonary Edema (Acute Respiratory Distress Syndrome) This syndrome is caused by increased pulmonary capillary permeability. Patients should be positioned with the head of the bed at a 45° elevation and should be given oxygen and IV diuretics. Positive-pressure ventilation should be started early if the immediate measures fail (**Chap. 305**). Rarely, patients may require extracorporeal membrane oxygenation.

Hypoglycemia An initial slow injection of 20% dextrose (2 mL/kg over 10 min) should be followed by an infusion of 10% dextrose (0.10 g/kg per hour). The blood glucose level should be checked regularly thereafter as recurrent hypoglycemia is common, particularly among patients receiving quinine. In severely ill patients, hypoglycemia commonly occurs together with metabolic (lactic) acidosis and carries a poor prognosis.

Sepsis Hypoglycemia or gram-negative septicemia should be suspected when the condition of any patient suddenly deteriorates for no obvious reason during antimalarial treatment. In malaria-endemic areas where a high proportion of children are parasitemic, it is usually impossible to distinguish severe malaria from bacterial sepsis with confidence. These children should be treated with both antimalarials and broad-spectrum antibiotics with activity against nontyphoidal *Salmonella* species from the outset. Empirical antibiotics should also be given to adults with >20% parasitemia. Antibiotics should be considered for severely ill patients of any age who are not responding to antimalarial treatment or deteriorate unexpectedly.

Other Complications Patients who develop spontaneous bleeding should be given fresh blood and IV vitamin K. Convulsions should be treated with IV or rectal benzodiazepines and, if necessary, respiratory support. Aspiration pneumonia should be suspected in any unconscious patient with convulsions, particularly with persistent hyperventilation; IV antimicrobial agents and oxygen should be administered, and pulmonary toilet should be undertaken.

GLOBAL CONSIDERATIONS

The goal of global eradication of malaria remains a challenge. Success will require strong leadership, increased national commitment, and substantial international support. The two main tools used to control malaria are insecticide-treated bed nets (ITNs), previously shown to reduce all-cause mortality in African children by 20%, and the ACTs. New drugs are in development. One vaccine (the RTS,S/AS01 vaccine) which, in African children, provided 35–40% protection against falciparum malaria over 4 years of follow-up, has recently been recommended by WHO for widespread roll-out. Challenges to malaria eradication include the widespread distribution of *Anopheles* breeding sites, the enormous number of infected persons, the emergence and spread of resistance in *P. falciparum* to ACTs, increasing insecticide resistance and behavioral changes (to avoid ITN contact) in anopheline mosquito vectors, and inadequacies in human and material resources, infrastructure, and control programs. Newer ITNs combine pyrethroids with piperonyl butoxide, which increases mosquito susceptibility to pyrethroids by inhibiting CYP P450. Eliminating vivax malaria is further hindered by the lack of a simple, safe, radical curative regimen.

MALARIA PREVENTION

Malaria may be contained by judicious use of insecticides to kill the mosquito vector, rapid diagnosis, patient management, and—where effective and feasible—administration of intermittent preventive treatments, seasonal malaria chemoprevention, or chemoprophylaxis to high-risk groups such as pregnant women and young children. Focal elimination of *P. falciparum* can be accelerated safely by mass treatment with slowly eliminated antimalarials such as

1734 dihydroartemisinin-piperaquine. Despite the enormous investment in efforts to develop a malaria vaccine, no safe, highly effective, long-lasting vaccine is likely to be available for general use in the near future. The licensed recombinant protein sporozoite-targeted adjuvanted vaccine RTS,S/AS01 was only moderately efficacious in protecting African children from malaria in field trials, and protection wanes rapidly. The vaccine is being deployed in Ghana, Kenya, and Malawi as part of a large-scale pilot project and has just been approved by WHO for general deployment. An irradiated live sporozoite vaccine is in late-stage development, and research is ongoing to develop a vaccine to protect against placental malaria (targeting VAR2CSA). While there is great promise for one or several malaria vaccines on the more distant horizon, prevention and control measures will continue to rely on antivector and drug-use strategies for the foreseeable future.

■ PERSONAL PROTECTION AGAINST MALARIA

Simple measures to reduce the frequency of bites by infected mosquitoes in malarious areas are very important. These measures include the avoidance of exposure to mosquitoes at their peak feeding times (usually dusk to dawn) and the use of insect repellents containing 10–35% DEET (or, if DEET is unacceptable, 7% picaridin), suitable clothing, and ITNs or other insecticide-impregnated materials. Widespread use of bed nets treated with residual pyrethroids reduces the incidence of malaria in areas where vectors bite indoors at night.

■ CHEMOPROPHYLAXIS

(Table 224-8; <https://wwwnc.cdc.gov/travel/yellowbook/2020/travel-related-infectious-diseases/malaria>) Recommendations for malaria prophylaxis depend on knowledge of local patterns of drug sensitivity in

TABLE 224-8 Drugs Used in the Prophylaxis of Malaria^a

DRUG	USAGE	ADULT DOSE	PEDIATRIC DOSE	COMMENTS
Atovaquone-proguanil (Malarone)	Prophylaxis in areas with chloroquine- or mefloquine-resistant <i>Plasmodium falciparum</i>	1 adult tablet PO ^b	5–8 kg: ½ pediatric tablet ^c daily ≥8–10 kg: ¾ pediatric tablet daily ≥10–20 kg: 1 pediatric tablet daily ≥20–30 kg: 2 pediatric tablets daily ≥30–40 kg: 3 pediatric tablets daily ≥40 kg: 1 adult tablet daily	Begin 1–2 days before travel to malarious areas. Take daily at the same time each day while in the malarious areas and for 7 days after leaving such areas. Atovaquone-proguanil is contraindicated in persons with severe renal impairment (creatinine clearance rate, <30 mL/min). In the absence of data, it is not recommended for children weighing <5 kg, pregnant women, or women breast-feeding infants weighing <5 kg. Atovaquone-proguanil should be taken with food or a milky drink.
Chloroquine phosphate (Aralen and generic)	Prophylaxis only in the very few areas with chloroquine-sensitive <i>P. falciparum</i> ^c or areas with <i>P. vivax</i> only	300 mg of base (500 mg of salt) PO once weekly	5 mg of base/kg (8.3 mg of salt/kg) PO once weekly, up to maximum adult dose of 300 mg of base	Begin 1–2 weeks before travel to malarious areas. Take weekly on the same day of the week while in the malarious areas and for 4 weeks after leaving such areas. Chloroquine may exacerbate psoriasis.
Doxycycline (many brand names and generic)	Prophylaxis in areas with chloroquine- or mefloquine-resistant <i>P. falciparum</i> ^d	100 mg PO qd (except in pregnant women; see Comments)	≥8 years of age: 2 mg/kg PO qd, up to adult dose	Begin 1–2 days before travel to malarious areas. Take daily at the same time each day while in the malarious areas and for 4 weeks after leaving such areas. Doxycycline is contraindicated in children aged <8 years and in pregnant women.
Hydroxychloroquine sulfate (Plaquenil)	An alternative to chloroquine for primary prophylaxis only in the very few areas with chloroquine-sensitive <i>P. falciparum</i> ^d or areas with <i>P. vivax</i> only	310 mg of base (400 mg of salt) PO once weekly	5 mg of base/kg (6.5 mg of salt/kg) PO once weekly, up to maximum adult dose of 310 mg of base	Begin 1–2 weeks before travel to malarious areas. Take weekly on the same day of the week while in the malarious areas and for 4 weeks after leaving such areas. Hydroxychloroquine may exacerbate psoriasis.
Mefloquine (Lariam and generic)	Prophylaxis in areas with chloroquine-resistant <i>P. falciparum</i> ^d	228 mg of base (250 mg of salt) PO once weekly	≤9 kg: 4.6 mg of base/kg (5 mg of salt/kg) PO once weekly 10–19 kg: ¼ tablet ^e once weekly 20–30 kg: ½ tablet once weekly 31–45 kg: ¾ tablet once weekly ≥46 kg: 1 tablet once weekly	Begin 1–2 weeks before travel to malarious areas. Take weekly on the same day of the week while in the malarious areas and for 4 weeks after leaving such areas. Mefloquine is contraindicated in persons allergic to this drug or related compounds (e.g., quinine and quinidine) and in persons with active or recent depression, generalized anxiety disorder, psychosis, schizophrenia, other major psychiatric disorders, or seizures. Use with caution in persons with psychiatric disturbances or a history of depression. Mefloquine is not recommended for persons with cardiac conduction abnormalities.
Primaquine	For prevention of malaria in areas with mainly <i>P. vivax</i>	30 mg of base (52.6 mg of salt) PO qd	0.5 mg of base/kg (0.8 mg of salt/kg) PO qd, up to adult dose; should be taken with food	Begin 1–2 days before travel to malarious areas. Take daily at the same time each day while in the malarious areas and for 7 days after leaving such areas. Primaquine is contraindicated in persons with G6PD deficiency. It is also contraindicated during pregnancy.
Primaquine	Used for presumptive antirelapse therapy (terminal prophylaxis) to decrease risk of relapses of <i>P. vivax</i> and <i>P. ovale</i>	30 mg of base (52.6 mg of salt) PO qd for 14 days after departure from the malarious area	0.5 mg of base/kg (0.8 mg of salt/kg), up to adult dose, PO qd for 14 days after departure from the malarious area	This therapy is indicated for persons who have had prolonged exposure to <i>P. vivax</i> and/or <i>P. ovale</i> . It is contraindicated in persons with G6PD deficiency as well as during pregnancy.

^aSeveral antimalarial drugs are formulated as different salts (e.g., phosphate, sulfate, hydrochloride) and are therefore prescribed as base equivalents. For example, chloroquine phosphate 250 salt contains 155 mg base equivalent. It is very important to check when prescribing that the correct dose is being given. ^bAn adult tablet contains 250 mg of atovaquone and 100 mg of proguanil hydrochloride. ^cA pediatric tablet contains 62.5 mg of atovaquone and 25 mg of proguanil hydrochloride. ^dVery few areas now have chloroquine-sensitive *falciparum* malaria. ^eOne tablet contains 228 mg of base (250 mg of salt).

Abbreviation: G6PD, glucose-6-phosphate dehydrogenase.

Source: Centers for Disease Control and Prevention, <https://www.cdc.gov/malaria/travelers/index.html>.

Plasmodium species and the likelihood of acquiring malarial infection. Drugs effective against resistant *P. falciparum* should be used (atovaquone-proguanil [Malarone], doxycycline, or mefloquine). Chemoprophylaxis is never entirely reliable, and malaria should always be considered in the differential diagnosis of fever in patients who have traveled to endemic areas, even if they are taking prophylactic antimalarial drugs.

Pregnant women planning to visit malarious areas should be warned about the potential risks and advised to avoid all nonessential travel. All pregnant women who live in endemic areas should be encouraged to attend regular antenatal clinics. Mefloquine is the only drug advised for pregnant women traveling to areas with drug-resistant malaria; this drug is generally considered safe in the second and third trimesters of pregnancy; the data on first-trimester exposure, although limited, are reassuring. Chloroquine and proguanil are regarded as safe, but there are now very few regions where these drugs can be recommended for protection. The safety of other prophylactic antimalarial agents in pregnancy has not been established. Antimalarial prophylaxis has been shown to reduce mortality rates among children between the ages of 3 months and 4 years in malaria-endemic areas; however, it is not a logistically or economically feasible option in many countries. The alternative—to give intermittent preventive treatment (IPT) to pregnant women, and in some areas to infants as well, or seasonal malaria chemoprevention (SMC) to young children—is being implemented. Other strategies are being evaluated, such as intermittent screening and treatment.

IPT in pregnancy (IPTp) involves giving treatment doses of sulfadoxine-pyrimethamine at each antenatal visit (maximum, once monthly) in the second and third trimesters of pregnancy. Women with HIV infection who are taking trimethoprim-sulfamethoxazole as prophylaxis should not be given concomitant sulfadoxine-pyrimethamine. Dihydroartemisinin-piperaquine is being evaluated as an alternative. IPT in infancy (IPTi) involves giving treatment doses of sulfadoxine-pyrimethamine along with the immunizations included in the WHO's Expanded Program on Immunization at 2, 3, and 9 months of life. Seasonal malaria chemoprevention involves giving monthly treatment doses of amodiaquine and sulfadoxine-pyrimethamine to children aged between 3 and 59 months during the 3- to 4-month rainy season across the Sahel region of Africa. Children born to nonimmune mothers in malaria-endemic areas (usually expatriates moving to these areas) should receive prophylaxis from birth.

Travelers to a malaria endemic region should start taking antimalarial drugs 2 days to 2 weeks before departure so that any untoward reactions can be detected before travel and so that therapeutic antimalarial blood concentrations will be present if and when any infections develop (Table 224-8). Antimalarial prophylaxis should continue for 4 weeks after the traveler has left the endemic area, except if atovaquone-proguanil or primaquine has been taken; these drugs have significant activities against the liver stage of the infection (causal prophylaxis) and can be discontinued 1 week after departure from the endemic area. If suspected malaria develops while a traveler is abroad, obtaining a reliable diagnosis and antimalarial treatment locally is a top priority. Presumptive self-treatment for malaria with atovaquone-proguanil (for 3 consecutive days) or one of the artemisinin-based combinations can be considered under special circumstances; medical advice on self-treatment should be sought before departure for malaria-endemic areas and as soon as possible after illness begins. Every effort should be made to confirm the diagnosis.

Atovaquone-proguanil (Malarone; 3.75/1.5 mg/kg or 250/100 mg, daily adult dose) is a fixed-combination, once-daily prophylactic agent that is very well tolerated by adults and children. This combination is effective against all types of malaria, including multidrug-resistant *falciparum* malaria. Atovaquone-proguanil is best taken with food or a milky drink to optimize absorption. It is not recommended if the estimated glomerular filtration rate is <30 mL/min. There are insufficient data on safety in pregnancy.

Mefloquine (250 mg of salt weekly, adult dose) has been widely used for malarial prophylaxis because it is usually effective against

multidrug-resistant *falciparum* malaria and is reasonably well tolerated. Mefloquine has been associated with rare episodes of psychosis and seizures at prophylactic doses; these reactions are more frequent at the higher doses used for treatment. More common side effects with prophylactic doses of mefloquine include mild nausea, dizziness, fuzzy thinking, disturbed sleep patterns, vivid dreams, dysphoria, and malaise. Mefloquine is contraindicated for use by travelers with known hypersensitivity and by persons with active or recent depression, anxiety disorder, psychosis, schizophrenia, another major psychiatric disorder, or seizures; it is not recommended for persons with cardiac conduction abnormalities, although the evidence that it is cardiotoxic is very weak. Confidence is increasing with regard to the safety of mefloquine prophylaxis during pregnancy; in studies in Africa, mefloquine prophylaxis was found to be effective and safe during pregnancy. Daily administration of doxycycline (100 mg daily, adult dose) is an effective alternative to atovaquone-proguanil or mefloquine. Doxycycline is generally well tolerated but may cause vulvovaginal thrush, diarrhea, and photosensitivity and is not recommended for prophylaxis in children <8 years old or pregnant women, although, evidence that it is harmful is lacking.

Chloroquine can no longer be relied upon to prevent *P. falciparum* infections in nearly all endemic areas but is still used to prevent and treat malaria due to the other human *Plasmodium* species and for *P. falciparum* malaria in Central American countries west and north of the Panama Canal and in Caribbean countries. Chloroquine-resistant *P. vivax* has been reported from parts of eastern Asia, Oceania, and Central and South America. High-level resistance in *P. vivax* is prevalent in Oceania and Indonesia. Chloroquine is generally well tolerated, although some patients cannot take it because of malaise, headache, visual symptoms (due to reversible keratopathy), gastrointestinal intolerance, alopecia, or pruritus. Chloroquine is considered safe in pregnancy. With chronic administration for >5 years, a characteristic dose-related retinopathy may develop, but this condition is rare at the doses used for antimalarial prophylaxis. Idiosyncratic or allergic reactions are also rare. Skeletal and/or cardiac myopathy is a potential problem with protracted prophylactic use, although it is more likely to occur at the high doses used in the treatment of rheumatoid arthritis. Neuropsychiatric reactions and skin rashes are unusual. Amodiaquine should not be used for weekly prophylaxis because continuous weekly use is associated with a high risk of agranulocytosis (~1 person in 2000) and hepatotoxicity (~1 person in 16,000). Chloroquine, amodiaquine, and piperaquine all cause moderate electrocardiograph QT prolongation but have not been associated with ventricular arrhythmias at therapeutic doses.

Primaquine (0.5 mg of base/kg or a daily adult dose of 30 mg taken with food), an 8-aminoquinoline compound, has proved safe and effective in the prevention of drug-resistant *falciparum* and *vivax* malaria in adults. Primaquine can be considered for adults (with the exception of pregnant women) who are intolerant to other recommended drugs. Abdominal pain can be prevented by taking primaquine with food. G6PD deficiency must be excluded before primaquine is prescribed. In the past, the dihydrofolate reductase inhibitors pyrimethamine and proguanil (chloroguanide) were administered widely, but the rapid selection of resistance in both *P. falciparum* and *P. vivax* has limited their use. Whereas antimalarial quinolines such as chloroquine (a 4-aminoquinoline) act only on the erythrocyte stage of parasitic development, the dihydrofolate reductase inhibitors (as well as atovaquone and primaquine) also inhibit preerythrocytic growth in the liver (causal prophylaxis) and development in the mosquito (sporontocidal activity). Proguanil is safe and well tolerated, although mouth ulceration occurs in ~8% of persons using this drug; it is considered safe for antimalarial prophylaxis in pregnancy. Prophylactic use of the combination of pyrimethamine and sulfadoxine is not recommended for weekly administration because of an unacceptable incidence of severe toxicity, principally exfoliative dermatitis and other skin rashes, agranulocytosis, hepatitis, and pulmonary eosinophilia (incidence, 1 in 7000; fatal reactions, 1 in 18,000).

Because of the increasing spread and intensity of antimalarial drug resistance (Fig. 224-10), the CDC recommends that travelers and

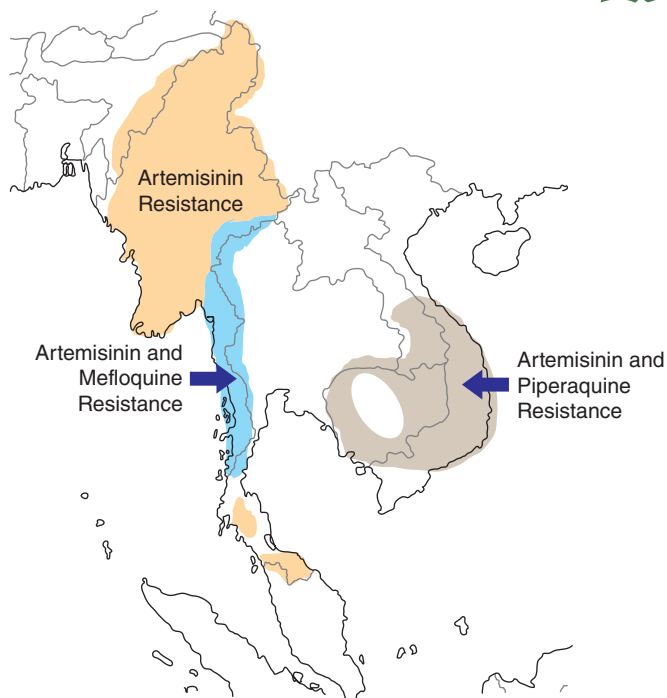


FIGURE 224-10 Current geographic extent of artemisinin resistance and artemisinin-based combination therapy partner drug resistance in *Plasmodium falciparum* in the Greater Mekong subregion.

their providers consider their destination, type of travel, and current medications and health risks when choosing antimalarial chemoprophylaxis. There is an increasingly appreciated problem of falsified and substandard antimalarial drugs (and other medicines) on the shelves of pharmacies in Southeast Asia and sub-Saharan Africa; hence, travelers should purchase their preventive drugs from a reputable source before going to a malarious country. Consultation for the evaluation of prophylaxis failures or treatment of malaria can be obtained from state and local health departments and the CDC Malaria Hotline (770-488-7788) or the CDC Emergency Operations Center (770-488-7100).

ACKNOWLEDGMENT

The authors gratefully acknowledge the substantial contributions of Joel G. Breman to this chapter in previous editions.

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Babesiosis

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Babesiosis is an emerging infectious disease caused by protozoan parasites of the genus *Babesia* that invade and eventually lyse red blood cells (RBCs). Most cases occur in the United States during the summer months and are caused by *Babesia microti*, a species typically found in small rodents and transmitted by the deer tick, *Ixodes scapularis*. Symptoms are those of a flu-like illness. A single standard course of atovaquone plus azithromycin often is sufficient to achieve cure. Highly immunocompromised patients are at risk of persistent infection and antibiotic resistance and should be treated for a longer duration than immunocompetent patients. Adjunct exchange transfusion can be useful for severe cases. In the absence of vaccine and prophylaxis, persons at risk of severe babesiosis should minimize their exposure to ticks and, if possible, avoid endemic areas.

ETIOLOGY, EPIDEMIOLOGY, AND MODES OF TRANSMISSION

A few *Babesia* species have been implicated as etiologic agents of human babesiosis. These species use wild or domesticated mammals as reservoir hosts and are maintained in their enzootic cycle by ticks. Humans are incidental, dead-end hosts. Most cases of human babesiosis are reported from across the Northern Hemisphere, but the predominant etiologic agent varies by continent.

United States • GEOGRAPHIC DISTRIBUTION *B. microti* is the etiologic agent of nearly all cases in the United States. Most cases (~95%) are reported from the Northeast and the upper Midwest, particularly from seven states: Massachusetts, Rhode Island, Connecticut, New York, New Jersey, Minnesota, and Wisconsin (Fig. 225-1). Aside from *B. microti*, other *Babesia* species seldom cause disease in the United States. Symptomatic infection with *Babesia duncani* and *B. duncani*-type organisms has been reported from Washington State, Oregon and California. Symptomatic infection with *Babesia divergens*-like organisms has been documented in Washington State but also in the central states of Arkansas, Missouri, Michigan, and Kentucky.

Incidence Babesiosis has been a nationally notifiable disease since January 2011. Cases are reported weekly by state health departments to the Centers for Disease Control and Prevention (CDC) via the National Notifiable Diseases Surveillance System. In 2018, a total of 2161 cases were reported from 28 of the 40 states in which babesiosis was notifiable. In 2008, when reporting to the CDC was yet to be instituted, 624 cases were made known to health departments in the seven highly endemic states; a decade earlier, this number was 230. The increase in incidence over time is best explained by a greater density of ticks in highly endemic areas as well as the steady expansion of the geographic range occupied by these ticks. Although *B. microti* and *Borrelia burgdorferi* (the agent of Lyme disease; Chap. 186) are maintained in their enzootic life cycle by the deer tick *I. scapularis*, the geographic expansion of babesiosis has lagged behind that of Lyme disease. This delay in geographic expansion likely reflects the poor ecologic fitness of *B. microti* compared with that of *B. burgdorferi* and is consistent with the observation that *B. burgdorferi* helps maintain *B. microti* in its enzootic cycle. Other factors contributing to the rise in incidence include greater exposure of residents to ticks due to forest fragmentation in suburban areas and more time spent in leisure activities in grassy or wooded areas. The seroprevalence of antibodies specific for *B. microti* has been as high as 9% among residents of highly endemic states, particularly in areas with the highest incidence of disease. Similar rates have been noted among blood donors in these highly endemic states, indicating that babesiosis is underreported and/or that asymptomatic infection is more common than is recognized.